TECHNICAL WHITEPAPER · 2026-08-20

GumBall6900

A signal-directed onchain portfolio acquisition and redemption protocol

The complete technical specification: exact formulas, state-transition tables, accounting identities, security invariants, precision analysis, threat model, and verification evidence.

VERSION

2.0.0

DATE

2026-08-20

SOURCE COMMIT

dc67d7c4d634…

PROTOCOL STATUS

Development snapshot at the pinned commit. Describes the current core after ADR 0033, ADR 0034, and ADR 0035; not approved for user funds.

DEPLOYMENT

Not deployed on any network. No signed deployment manifest exists.

INDEPENDENT AUDIT

No independent external audit has been performed.

Contents

Safety status. The protocol is **not deployed, not independently audited, and not approved for user funds.** Every claim in this edition was verified against the Solidity tree at commit `dc67d7c`. A local green build is engineering evidence, never a safety, audit, or release claim.

Governance is unselected. The core contains no `ProtocolGovernor`, `TimelockController`, generic executor, or provider-specific governance adapter; ADR 0034 removed them. `Resonance` is the only owned core contract, and the external governance system that will own it has not been selected. §15 and §27 specify exactly what the core does and does not guarantee as a consequence.

Companion sources. The compact whitepaper and one-page sheet are built from `docs/whitepaper/` and `docs/one-pager/gumball6900/` via `pnpm docs:whitepaper` and `pnpm docs:one-pager`. This long-form source builds to `output/pdf/GumBall6900-whitepaper.pdf` via `pnpm docs:longform`.

1. Abstract

GUM BALL 6900 is an immutable, governance-minimized protocol that converts recurring onchain revenue into a permissionlessly redeemable portfolio of arbitrary ERC-20 assets, with allocation directed by non-transferable staked governance weight rather than by a manager, an oracle, or an index methodology.

The protocol issues a token, **GBX**, through a multislot mining market in which slot occupancy is sold by hourly descending-price auctions denominated in an external stablecoin, **USDG**. Eighty percent of a replacement payment compensates the displaced occupant; the remainder, together with fees from a permanently locked Uniswap v4 GBX/USDG position, constitutes protocol revenue.

Revenue is placed into a single rolling seven-day emission schedule held by **Resonance** and allocated continuously across **Strategies** in proportion to the non-transferable staked weight (**SignalGBX**, ticker sGBX) allocated to each Strategy during each elapsed interval. A Strategy is a bounded descending-price auction that exchanges its accumulated USDG for a fixed target asset. Every auction payment is classified by a bounded, cumulatively exact rule: the signaler share is a single global parameter defaulting to 10% and capped at 20% (ADR 0036), and the remainder becomes an irrevocable liability to **Fund**, an ownerless treasury with no asset registry and no administrative surface. At least 80% of every acquisition therefore reaches Fund regardless of who holds the owner address.

GBX holders redeem by burning GBX and nominating an arbitrary set of unique non-GBX token addresses, receiving for each the floored pro-rata share of Fund's balance against a single effective pre-burn supply snapshot that includes all accrued unminted mining. Signaler compensation has two sources: the automatic 10% acquisition share, and **Bribes**, permissionlessly funded reward streams attached to each Strategy, capped at eight reward tokens.

Protocol administration is reduced to three owner-gated calls on a single contract: `Resonance.addStrategy`, `Resonance.killStrategy`, and `Resonance.addBribeReward`. Every other core contract is ownerless or has consumed its one-time binding. The core implements no proposal, quorum, voting, delay, or cancellation semantics of its own; ADR 0034 removed the in-repository Governor and Timelock in favour of an external governance system that has not yet been selected. `SignalGBX` retains non-transferable ERC20Votes checkpoints for that future integration, and the core assigns them no meaning. Selection, review, and the ownership handoff that removes the temporary deployment owner remain deployment blockers.

This document specifies the implemented mathematics exactly, states the accounting identities the implementation can actually prove (and explicitly declines to assert those it cannot), enumerates state transitions, and presents the threat model and residual risks.

2. Motivation

Pooled onchain asset vehicles have converged on a small set of trust concessions that are, individually, defensible and collectively fatal to the claim of trust minimization:

1. **Discretionary allocation.** A manager, multisig, or unbounded DAO decides what the vehicle holds. Holders trust both competence and honesty.
2. **Upgradeability.** A proxy admin can change redemption arithmetic after deposits are taken.
3. **Pausability.** An emergency actor can suspend exit while entry or accrual continues.
4. **Oracle dependence.** A price feed determines mint and redemption ratios, importing the oracle's failure modes and manipulation surface into the vehicle's core accounting.
5. **Registry curation.** An asset allow-list is maintained by a privileged party, so a single broken or malicious entry can block redemption for every other asset.

Each concession exists for a reason. Removing them requires replacing the function they served with a mechanism that does not require trust.

GUM BALL 6900 makes five substitutions:

CONCESSION REMOVED	REPLACEMENT MECHANISM
Discretionary allocation	Continuous, revocable, per-Strategy stake allocation (“signal”) that directs revenue by weight
Upgradeability	Immutable, non-proxied deployment with reciprocal one-time bindings
Pausability	Unpausable redemption; failure isolation by caller-selected asset baskets and deferred liabilities
Oracle pricing	Bounded descending-price auctions in which the clearing fill *is* price discovery
Registry curation	Registry-free raw-token treasury; the redeemer nominates the assets they wish to receive

The design accepts specific, enumerated costs for these substitutions — permanent dust accumulation, unrecoverable deployment errors, unbounded reward abandonment in retired Strategies, and the absence of any emergency response. These are stated in §38 and §39 rather than minimized.

3. Design goals

- **G1 — Immutability.** No contract may be upgraded, paused, migrated, or administratively drained. When a design choice trades governance flexibility against immutability, immutability wins and the consequence is recorded.
- **G2 — Minimal continuing authority.** The permanent administrative surface must be enumerable, selector-bounded, and small enough to state in one sentence.
- **G3 — Oracle independence.** No protocol accounting may depend on an external price, NAV, or valuation.
- **G4 — Exit liveness.** Redemption and stake withdrawal must never depend on the cooperation, solvency, or correctness of any third-party token other than the one being withdrawn.

- **G5 — Failure isolation.** A malformed, frozen, or malicious token must be able to block only its own payout, never another party's.
- **G6 — Exact value movement.** Every value transfer must verify exact sender debit and receiver credit, failing closed on inexact movement rather than silently absorbing loss.
- **G7 — Bounded work.** Every mandatory loop — reward tokens, mining slots, redemption baskets — must be bounded by a code constant or by caller-supplied input the caller pays for.
- **G8 — No retroactive redirection.** A weight change must never redirect value that accrued under prior weights.
- **G9 — Explicit accounting.** Where exact conservation is achievable it must be proven by identity; where it is not achievable, the residue must be classified and disclosed rather than implied to be zero.

4. Non-goals

- **N1 — No index methodology.** The set of Strategies registered in `Resonance` constitutes the protocol's index membership: it is the curated list of assets the protocol targets. What the protocol does **not** provide is index methodology — there is no target weighting, rebalancing, drift correction, reconstitution rule, or NAV. Fund holds whatever Strategies acquired plus whatever anyone sent it, in whatever proportions resulted. Membership is defined by registration in `Resonance`, never by a Fund balance (§24.2).
- **N2 — Not a valuation system.** The protocol never computes NAV, backing-per-token, or asset prices onchain.
- **N3 — Not a yield product.** No return, distribution, or performance is promised or engineered.
- **N4 — Not a general DAO.** The core implements no voting, proposal, or execution machinery at all. Protocol administration is three owner-gated calls on one contract (§15.1).
- **N5 — Not fee-on-transfer or rebase compatible.** Exact-delta checks make such tokens revert; this is fail-closed evidence, not support.
- **N6 — No emergency response.** There is deliberately no guardian, veto, circuit breaker, or recovery path.
- **N7 — No keeper infrastructure.** No function pays a caller bounty. All maintenance is voluntary and permissionless.
- **N8 — No dust conservation guarantee in Resonance.** Bribe conserves sub-unit carry exactly; Resonance deliberately does not, and no lifetime bound on its residue is claimed.

5. Terminology

TERM	DEFINITION
GBX	Transferable ERC-20 with ERC-2612 permit, 18 decimals. No vote checkpoints. Mined, staked, burned.
sGBX / SignalGBX	Non-transferable ERC-20 with ERC20Votes, 18 decimals. Minted 1:1 against staked GBX.
USDG	External stablecoin used for revenue. Six decimals **by deployment assumption**, not by code enforcement.
Signal	An absolute quantity of sGBX a specific account has allocated to a specific Strategy.
Allocated balance	The aggregate sGBX an account has committed across all live and killed Strategies; not withdrawable.
Strategy	A bounded descending-price auction exchanging accumulated USDG for one fixed payment token.
Live / killed Strategy	A Strategy accepting new signal and future revenue / one permanently excluded from both.

TERM	DEFINITION
Bribe	Per-Strategy multi-token reward stream, permissionlessly funded, paid to that Strategy's signalers.
BribeRouter	Per-Strategy contract converting an auction payment into an irrevocable Fund liability.
Fund	Ownerless raw-token treasury. Redemption and GBX burning are its only value exits.
Slot	One mining position accruing GBX at a tenure-locked rate; occupancy sold by hourly auction.
Epoch	One auction round, identified by a monotonically increasing `epochId` used for fill-race protection.
Reverse Dutch auction	The repository's term for the descending-price mechanism in `Mine` and `Strategy`. See §43 discrepancy D-1.
Reward period / stream	A fixed seven-day emission schedule with a base rate plus a front-loaded remainder.
Carry	Sub-unit reward precision retained across checkpoints rather than discarded.
Surplus	Value held by a contract that is not a liability to anyone and has no recovery path.
Checkpoint	Advancing lazily-accrued state to the current timestamp before mutating weights or balances.
Resonance owner	The single address holding the four continuing administration capabilities; intended to become an external governance executor (§15).

5.1 Notation

Throughout, `⌊x⌋` denotes integer floor division as performed by the EVM. All quantities are raw integer token units unless explicitly stated otherwise. `1 GBX = 10^18` raw units. Under the intended deployment, `1 USDG = 10^6` raw units. Timestamps are `block.timestamp` in seconds. `mulDiv(a, b, c)` denotes OpenZeppelin's `Math.mulDiv`, which computes `⌊a·b/c⌋` at 512-bit intermediate precision and therefore cannot overflow on the intermediate product.

6. Actors

ACTOR	CAPABILITY	TRUSTED FOR
Miner	Pays USDG to occupy a slot; accrues GBX at a tenure-locked rate; claims displaced-miner payments.	Nothing
Staker	Stakes GBX for sGBX; holds voting weight; may keep sGBX idle.	Nothing
Signaler	Allocates sGBX to Strategies, directing revenue and earning Bribe rewards.	Nothing
Strategy buyer	Fills a Strategy auction, paying the target asset and receiving accumulated USDG.	Nothing
Bribe funder	Permissionlessly streams reward tokens to a Strategy's signalers.	Nothing
Redeemer	Burns GBX and nominates assets to withdraw pro rata from Fund.	Nothing
Harvester	Permissionlessly collects Uniswap v4 fees into the protocol.	Nothing
Checkpointer / router	Permissionlessly advances lazy state and forwards qualifying revenue.	Liveness only
Resonance owner	Adds Strategies, kills Strategies, and registers Bribe reward tokens (§15.1).	**Fully trusted for those three capabilities**
Deployment coordinator	Executes the one-time bindings, creates bootstrap Strategies, then renounces all authority.	**Fully trusted, once, irreversibly**

The deployment coordinator is the protocol's single unavoidable trusted party. Its authority is temporary by procedure rather than by code (§28.4), and every error it can make is permanent.

7. System overview

The protocol comprises eleven deployed contract types. The economic cycle has five stages.

Stage 1 — Issuance and revenue origination. `Mine` holds exactly sixteen permanent slots. Each slot continuously accrues GBX at a rate fixed for its occupant's tenure. Occupancy is transferred by paying the slot's current hourly descending price in USDG. On an occupied slot, `⌊price·8000/10000⌋` accrues as a pull claim for the displaced occupant and the remainder routes to `ResonanceRouter`; on an empty slot the whole payment routes. Independently, `LiquidityPosition` permanently holds one Uniswap v4 GBX/USDG position whose fees anyone may harvest: harvested USDG routes to `ResonanceRouter`, harvested GBX is transferred to `Fund` and burned in the same transaction.

Stage 2 — Revenue scheduling. `ResonanceRouter` withholds any nonzero balance smaller than the exact USDG remaining in Resonance's active schedule. Once its balance qualifies, it forwards its entire balance; `Resonance` checkpoints elapsed emission, combines the notification with the remainder, and restarts a seven-day schedule.

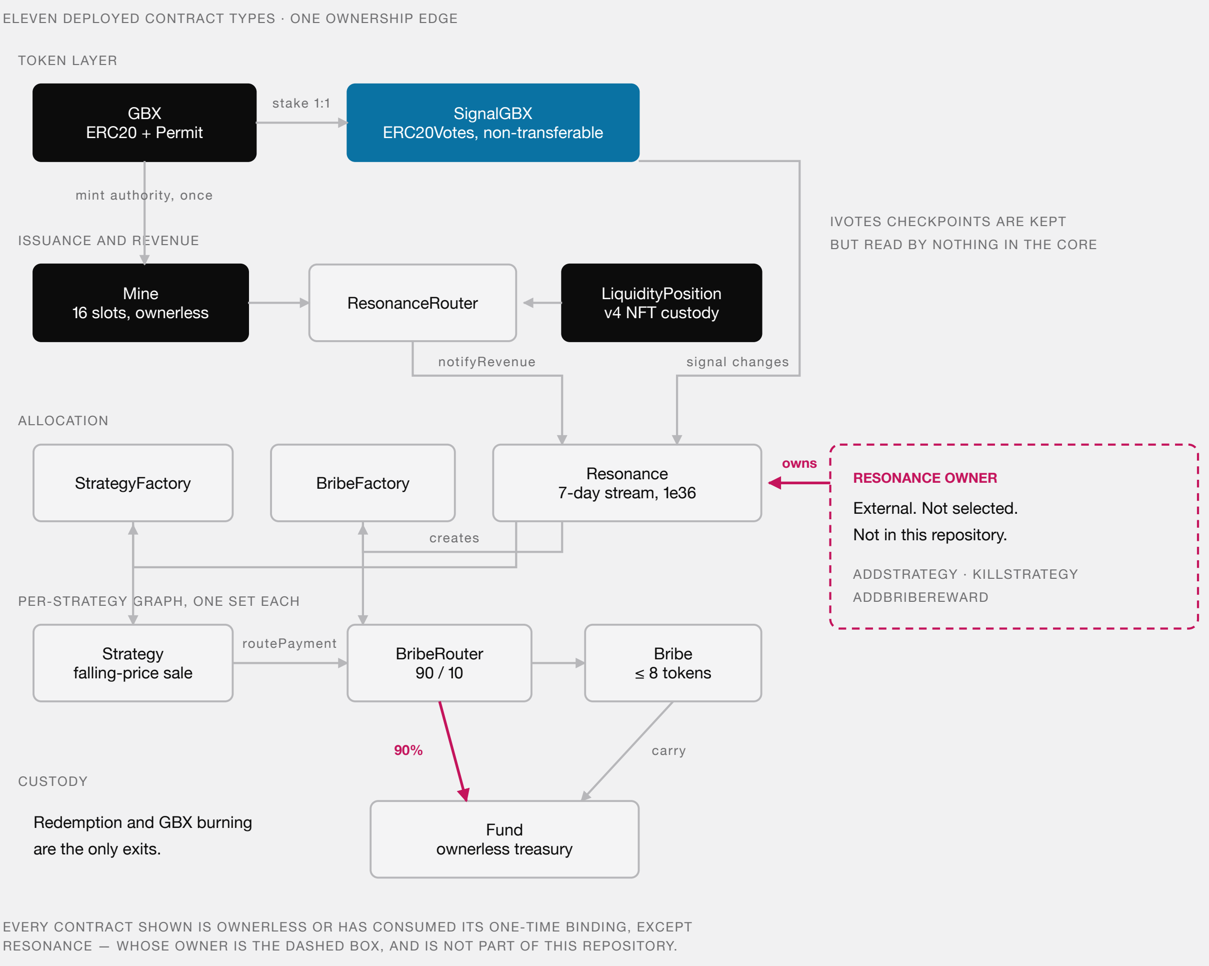
Stage 3 — Signal-weighted allocation. `SignalGBX` is the sole external signal coordinator. Its restricted hooks on `Resonance` checkpoint elapsed emission before mutating weights. Resonance maintains a Synthetix-shaped cumulative index at `1e36` precision over `totalSignalWeight`, the sum of recorded weights across *live* Strategies only.

Stage 4 — Acquisition. `Strategy.buy` first pulls the Strategy's released USDG from Resonance, then transfers its entire USDG balance to a buyer-nominated receiver in exchange for the current descending price in the Strategy's fixed payment token. That payment is routed through `BribeRouter` and classified 90% to an irrevocable `Fund` liability and 10% to the paired `Bribe`, each settled by its own separate permissionless call.

Stage 5 — Redemption. `Fund.redeem` checkpoints every mining slot, snapshots `GBX.totalSupply()` and its balance of each nominated token, pulls and burns the redeemer's GBX, and transfers each floored pro-rata share atomically.

Signaler compensation is delivered by `Bribe` contracts, funded both automatically by the 10% acquisition share and by anyone who chooses to add further rewards.

8. Contract graph



THE DEPLOYED CONTRACT GRAPH. ELEVEN CONTRACT TYPES, ONE OWNERSHIP EDGE, AND AN OWNER THAT IS DELIBERATELY DRAWN OUTSIDE THE SYSTEM BECAUSE ADR 0034 REMOVED IT FROM THIS REPOSITORY.

8.1 Cardinality

CONTRACT	INSTANCES
`GBX`	1
`SignalGBX`	1

CONTRACT	INSTANCES
`Mine`	1
`LiquidityPosition`	1
`ResonanceRouter`	1
`Resonance`	1
`StrategyFactory`	1
`BribeFactory`	1
`Fund`	1
`Strategy`	n, one per registered Strategy
`BribeRouter`	n, one per Strategy, deployed with it
`Bribe`	n, one per Strategy, deployed before it

9. Authority and ownership graph

NO OWNER
Nothing can be changed by anyone, ever.

Mine	sixteen slots, fixed
Fund	the treasury itself
LiquidityPosition	the v4 position
GBX	minter locked once
Strategy	auction parameters fixed
BribeRouter	90/10 hard-coded

ONE OWNER
Resonance, and only these three calls:

— add a Strategy
— retire a Strategy
— register a Bribe reward token

The holder of that address is **not yet chosen.**

There is no upgrade path, pause switch, sweep, or migration route anywhere in the protocol — so this map is the complete authority surface, not a summary of it.

THE COMPLETE AUTHORITY SURFACE. ONE CONTRACT HAS AN OWNER; THE REST CANNOT BE ALTERED BY ANYONE. WHO HOLDS THAT OWNER ADDRESS IS THE PROTOCOL’S LARGEST OPEN QUESTION.

9.1 Owned contracts and their permanent authority

CONTRACT	OWNER AFTER SETUP	CONTINUING OWNER-GATED FUNCTIONS
`Resonance`	External executor (**unselected**)	`addStrategy` , `killStrategy` , `addBribeReward` , plus inherited `transferOwnership` / `renounceOwnership`
`Mine`	**none**	none — sixteen slots are fixed at construction
`SignalGBX`	(setup owner)	none remaining — `setResonance` is consumed at deployment
`StrategyFactory`	(setup owner)	none remaining — `setResonance` is consumed at deployment
`BribeFactory`	(setup owner)	none remaining — `setResonance` is consumed at deployment
`GBX`	—	none — `setMinter` is single-use and permanently locks
`Fund`	**none**	none — contract is not `Ownable`
`LiquidityPosition`	**none**	none — contract is not `Ownable`
`Strategy`	**none**	none
`BribeRouter`	**none**	none
`Bribe`	**none**	`addRewardToken` , callable only by the immutable `Resonance`

`SignalGBX` , `StrategyFactory` , and `BribeFactory` retain a nominal `Ownable` owner after their one-time binding is consumed, but that owner has no remaining function to call. `Resonance.setResonanceRouter` is likewise single-use.

9.2 The four continuing administration actions

SELECTOR	TARGET	EFFECT	REVERSIBLE?
`Resonance.addStrategy`	`Resonance`	Deploys a Strategy, BribeRouter, and Bribe; registers payment token	No
`Resonance.killStrategy`	`Resonance`	Permanently excludes a Strategy from new signal and future revenue	**No**
`Resonance.addBribeReward`	`Resonance`	Appends a reward token to a Strategy's Bribe, within the cap of 8	**No**
`Resonance.setBribeBps`	`Resonance`	Sets the global signaler share of acquisitions, bounded `[0, 2000]`	Yes

The first three are irreversible. `addStrategy` is reversible only in the sense that the created Strategy can later be killed; the deployed contracts persist forever. `setBribeBps` is the sole reversible action and the sole economic parameter: it is bounded in code and applies prospectively, so it cannot reclassify an amount already settled.

9.3 Authority explicitly absent

No contract in `packages/contracts/src` contains a proxy, `Initializable` , `UUPSUpgradeable` , `Pausable` , a `delegatecall` , a sweep or rescue function, a successor binding, a migration routine, an emission setter, a fee setter, a price oracle, an entropy source, or a claim-redirection path.

There is also no governance machinery: no `Governor` , no `TimelockController` , no generic executor, and no provider-specific adapter. ADR 0034 removed them, so the guarantees such a stack would have supplied — proposal filtering, quorum, voting periods, execution delay, cancellation rules — are supplied by nothing in this repository. They become properties of whichever external system is later given ownership of `Resonance` (§15.2).

10. Token and asset-flow graph

10.1 USDG

ORIGIN	PATH	TERMINAL STATE
Miner replacement payment	payer → `Mine` ; `⌊p·8/10⌋` retained as claim, remainder → `ResonanceRouter`	Displaced-miner claim, or revenue
Uniswap v4 fees	`LiquidityPosition` → `ResonanceRouter`	Revenue
Revenue	`ResonanceRouter` → `Resonance` (on qualification) → `Strategy` (on distribute)	Auction inventory
Auction inventory	`Strategy` → buyer-nominated `revenueReceiver`	Exits the protocol
Rounding floors, zero-signal intervals, direct donations	remains in `Resonance`	**Permanent surplus, unrecoverable**
Direct donation to Router	remains in `ResonanceRouter` until a later qualifying route	Eventually revenue

10.2 GBX

ORIGIN	PATH	TERMINAL STATE
Genesis allocation	constructor → genesis recipient → Uniswap v4 position	Permanently locked in `LiquidityPosition`
Mining issuance	`Mine` mints to slot occupant	Circulating
Signal	holder → `SignalGBX` custody, sGBX minted 1:1, committed to a Strategy	Escrowed, redeemable by withdrawSignal
Direct donation to sGBX	remains in `SignalGBX`	**Stranded surplus, no receipt**
Harvested LP fees in GBX	`LiquidityPosition` → `Fund` → burned atomically	Destroyed
GBX-denominated auction	buyer → `Strategy` → `BribeRouter` → `Fund` , burnable by anyone	Destroyed when burned
Redemption	redeemer → `Fund` → burned	Destroyed

10.3 Acquired assets and Bribe reward tokens

ORIGIN	PATH	TERMINAL STATE
Auction payment (90%)	buyer → `Strategy` → `BribeRouter` → `Fund`	Fund backing until redeemed
Auction payment (10%)	buyer → `Strategy` → `BribeRouter` → paired `Bribe`	Streamed to that Strategy's signalers
Bribe funding	funder → `Bribe` → signalers	Signaler balances
Bribe carry classification	`Bribe` sub-unit remainders → `Fund`	Fund backing
Unsolicited transfer	anyone → `Fund`	Fund backing, unreviewed
Abandoned Bribe rewards	remains in `Bribe` after final signaler exit on a dead Strategy	**Permanently unclaimable, unbounded**

Structural invariant of the flow graph: the only paths *out* of `Fund` are `redeem` and `burnGBX` . There is no path from `Fund` to any administrator, and no caller-selectable destination for any share of an auction payment.

11. GBX

`GBX` is `ERC20, ERC20Permit`. Name `"GUM BALL 6900"`, symbol `"GBX"`, 18 decimals. It deliberately does **not** inherit `ERC20Votes`: governance weight exists only as sGBX (§13).

11.1 State

VARIABLE	TYPE	MEANING
`minter`	`address`	Current mint authority
`minterLocked`	`bool`	Whether the one-time Mine handoff has completed permanently
`lifetimeMinted`	`uint256`	Cumulative raw units created, including genesis
`lifetimeBurned`	`uint256`	Cumulative raw units destroyed

11.2 Genesis allocation

```
GENESIS_LIQUIDITY_ALLOCATION = 20_000_000 · 10^18 = 2·10^25 raw units
```

Minted once in the constructor to `genesisLiquidityRecipient`, with `lifetimeMinted` initialized to the same value. No other allocation, vesting schedule, treasury reserve, or airdrop exists in the token contract.

11.3 Supply identity

Identity I-1. At every block:

```
GBX.totalSupply() = GBX.lifetimeMinted() − GBX.lifetimeBurned()
```

Proof sketch. `lifetimeMinted` is incremented by exactly `amount` immediately before every `_mint(account, amount)` (constructor and `mint`), and `lifetimeBurned` by exactly `amount` immediately before every `_burn`. Both counters are monotonically non-decreasing and written nowhere else. ∎

Verified by `testFuzz_SupplyEqualsLifetimeMintedMinusBurned` and `invariant_GBXSupplyReconcilesWithBurns`.

There is no supply cap. Under the emission schedule of §12.5, cumulative issuance grows without bound in time.

11.4 Mint authority handoff

`setMinter(newMinter)` succeeds only when all of the following hold, and then permanently sets `minterLocked = true`:

```
msg.sender == minter
minterLocked == false
newMinter ≠ 0   ∧   newMinter ≠ minter   ∧   newMinter.code.length > 0
IMine(newMinter).gbx() == address(this)       -- reciprocal identity, try/catch guarded
```

`mint` requires both `msg.sender == minter` and `minterLocked == true`. Therefore the deployment-time coordinator can never mint, and after the handoff exactly one immutable address can mint forever. `burn` touches neither field, so burning cannot reopen authority.

Limitation. The reciprocal check proves the candidate *claims* the same GBX. It cannot distinguish a malicious lookalike returning the expected value. This is finding **M-03**, an open High release gate (§41.3).

12. Mining and issuance

`Mine` is an ownerless `ReentrancyGuard`. It is the sole GBX issuer after the handoff.

12.1 Immutable parameters and their bounds

PARAMETER	SYMBOL	CONSTRUCTOR BOUND	UNITS
`priceMultiplier`	`m`	`1.1·10^18 ≤ m ≤ 3·10^18`	1e18 fixed point
`minimumInitialPrice`	`P_min`	`10^6 ≤ P_min ≤ 2^192−1`	raw USDG
`initialTps`	`u_0`	`0 < u_0 ≤ 10^24`	raw GBX per second
`tailTps`	`u_∞`	`16 ≤ u_∞ ≤ u_0`	raw GBX per second
`halvingAmount`	`H`	`10^3·10^18 ≤ H ≤ 10^27`	raw GBX cumulative

Additionally `IRevenueRouterIdentity(resonanceRouter).usdg() == usdg` must hold.

The lower bound `u_∞ ≥ 16 = SLOT_COUNT` guarantees `⌊u_∞ / 16⌋ ≥ 1`, so a newly occupied slot always receives a strictly positive rate.

Constants: `BPS = 10_000`, `PREVIOUS_MINER_BPS = 8_000`, `PRICE_PRECISION = 10^18`, `PRICE_DECAY_PERIOD = 3600`, `SLOT_COUNT = 16`.

12.2 Slot state

```
struct Slot {
    uint256 epochId;          // monotonically increasing fill counter, starts at 1
    uint256 initialPrice;     // starting USDG price of the current auction
    uint256 auctionStartedAt; // timestamp the current auction began
    uint256 lastAccruedAt;    // start of this tenure's unsettled accrual
    uint256 tps;              // raw GBX per second, locked for this tenure
    address miner;            // zero when empty
}
```

An empty slot is `{epochId: 1, initialPrice: P_min, auctionStartedAt: now, lastAccruedAt: now, tps: 0, miner: 0}`.

12.3 Replacement price

Formula F-1 (slot price). For elapsed `e = now − auctionStartedAt` and `D = 3600`:

```
price(e) = initialPrice − ⌊initialPrice · e / D⌋     for e < D
price(e) = 0                                          for e ≥ D
```

Symbols. `initialPrice`, raw USDG units, `uint256` bounded by `2^192−1`; `e`, `D` in seconds. *Rounding.* The subtracted term floors, so `price(e)` lies at or **above** the exact real-valued line — rounding favors the protocol and the displaced miner, never the incoming payer. `price` is a non-increasing step function. *Overflow.* `Math.mulDiv` at 512-bit intermediate precision; with `initialPrice < 2^192` and `e < 2^256`, no overflow.

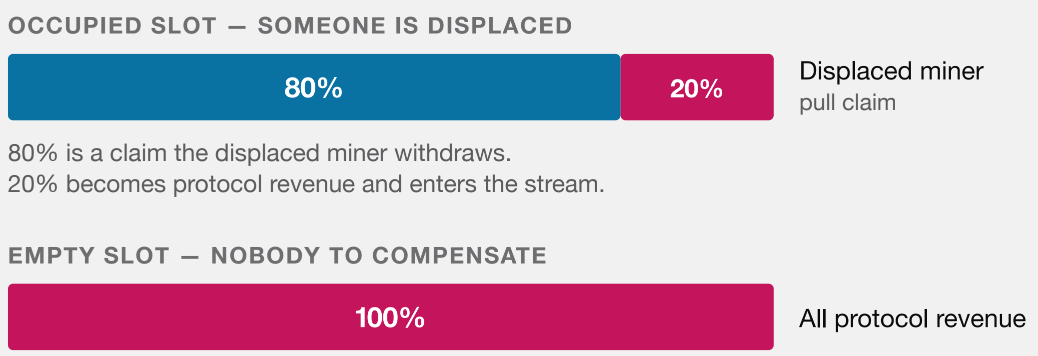
Worked example. `initialPrice = 2_000_000` (2.000000 USDG). At `e = 900`: `price = 2_000_000 − ⌊2_000_000·900/3600⌋ = 2_000_000 − 500_000 = 1_500_000`. At `e = 1800`, `1_000_000`; at `e = 2700`, `500_000`; at `e = 3600`, `0`. This is the curve reproduced in `packages/simulations/fixtures/economic-scenarios.json`.

12.4 Payment allocation

Formula F-2 (replacement split). For clearing price `p` and previous occupant `M`:

```
if p = 0:                        claim = 0,                     revenue = 0
if p > 0 and M = 0 (empty):      claim = 0,                     revenue = p
if p > 0 and M ≠ 0 (occupied):   claim = ⌊p · 8000 / 10000⌋,    revenue = p − claim
```

Rounding. `revenue = p − ⌊0.8p⌋ = ⌈0.2p⌉`. The rounding unit accrues to the **protocol**, not the displaced miner. *Dust.* At most 1 raw USDG unit per replacement, always in the protocol's favor. There is no accepted loss.



MINING HANDOFF. TAKING AN OCCUPIED SLOT COMPENSATES THE MINER YOU DISPLACED; TAKING AN EMPTY ONE FUNDS THE PROTOCOL ENTIRELY.

Worked example. `p = 1_000_000`: `claim = 800_000`, `revenue = 200_000`. `p = 1_000_003`: `claim = ⌊800_002.4⌋ = 800_002`, `revenue = 200_001`. Note `200_001 > 0.2·1_000_003 = 200_000.6`, confirming the ceiling behavior.

Identity I-2 (Mine solvency).

```
USDG.balanceOf(Mine) ≥ Mine.totalClaimable()
```

with equality when no unsolicited donation has been made. `claimable[a]` and `totalClaimable` are incremented together in `_allocatePayment` and decremented together in `claim`; routed revenue leaves the contract in the same transaction it arrives. Verified by `invariant_MineIsSolventAgainstReplacementClaims` and `testFuzz_MineRevenueAndHandoffClaimsReachFinalDestinationsWithoutDust`.

12.5 Global emission schedule

Formula F-3 (global rate). Let `T_k` be the cumulative-mining threshold triggering the `k`-th halving:

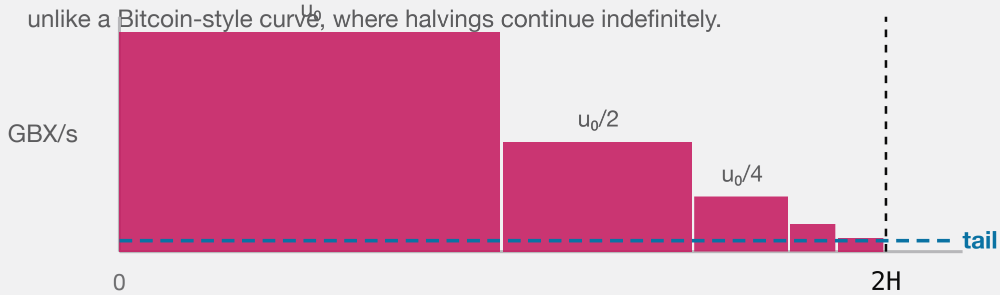
```
T_1 = H
T_{k+1} = T_k + ⌊H / 2^k⌋        (implemented as H >> k)
```

The global rate after `k` halvings is `u_k = ⌊u_0 / 2^k⌋` (implemented as `u_0 >> k`). The schedule terminates at the smallest `k*` with `u_{k*} ≤ u_∞`, after which the rate is permanently `u_∞` and the next threshold is set to `type(uint256).max`.

Symbols. `u_k` in raw GBX per second; `T_k`, `H` in raw GBX cumulative. **Overflow.** All operations are shifts and additions on `uint256`; `H ≤ 10^27` and `u_0 ≤ 10^24` keep every intermediate far below `2^256`.

GLOBAL RATE AGAINST CUMULATIVE MINING

The thresholds halve too, so the whole schedule finishes below 2H — unlike a Bitcoin-style curve, where halvings continue indefinitely.



After the last halving the rate is permanently the tail. Issuance never stops and never reaches zero.

ISSUANCE HALVES AT THRESHOLDS THAT THEMSELVES HALVE, SO THE ENTIRE SCHEDULE COMPLETES BELOW TWICE THE FIRST THRESHOLD AND THEN RUNS ON A PERMANENT TAIL.

Structural consequence. Because the thresholds themselves halve,

```
lim_{k→∞} T_k = Σ_{i≥0} ⌊H/2^i⌋ < 2H
```

so the **entire halving schedule completes before cumulative mining reaches `2H`**. After that point, global issuance is permanently `u_∞` raw GBX per second and never changes again. This is a materially different shape from a Bitcoin-style schedule, where thresholds are uniform and halvings continue indefinitely.

Worked example. Using the illustrative model values in `economic-scenarios.json`: `u_0` corresponding to 100 GBX/hour, `H = 490_000_000·10^18`. The first halving occurs at cumulative mined `490,000,000 GBX`, dropping the global rate to 50 GBX/hour; the second at `490,000,000 + 245,000,000 = 735,000,000 GBX`, and so on, converging below `980,000,000 GBX` cumulative. An incumbent's rate is unaffected by any of these crossings (§12.6).

> **These are illustrative model values, not production parameters.** The exact `u_0`, `H`, and `u_∞` have not been selected or independently modelled. This is finding **M-04**, an open High release gate.

12.6 Tenure-locked accrual

Formula F-4 (slot accrual). For an occupied slot,

```
pendingEmission(i) = (now − slots[i].lastAccruedAt) · slots[i].tps
pendingEmission()  = storedPendingEmission + (now − pendingUpdatedAt) · aggregateTps
effectiveTotalSupply = GBX.totalSupply() + pendingEmission()
```

`aggregateTps` is the sum of the sixteen tenure rates. A handoff first advances the system accumulator at that old aggregate rate, then mints only the outgoing slot's accrued amount and subtracts the same amount from stored pending emission. Unrelated slots are neither iterated nor mutated.

Formula F-5 (new tenure rate). On a fill, after syncing pending emission and settling the outgoing slot:

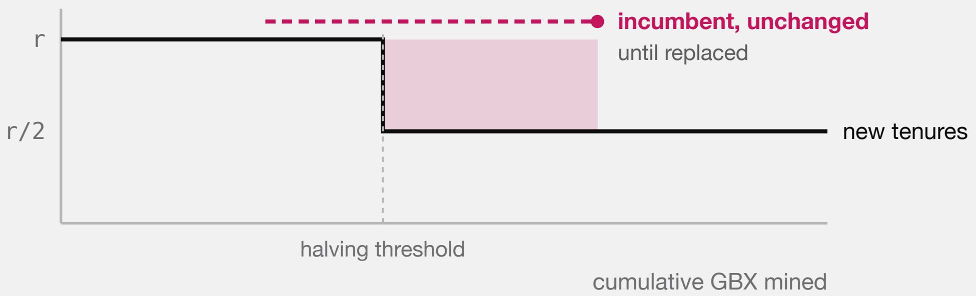
```
newSlot.tps = ⌊ globalTps(totalMined + storedPendingEmission) / 16 ⌋
```

Rounding. The division residue is **unissued** — it is never minted to anyone. This is a deliberate, permanent reduction in aggregate issuance relative to the undivided global rate, bounded by `15` raw units per second across all slots.

Invariant. `slot.tps` is assigned in exactly one place in the contract: the `Slot` struct literal inside `mine()`. No other handoff, threshold crossing, or redemption modifies it. Verified by `test_HalvingUsesEconomicAccrualAndNeverRepricesAnIncumbent`, `testFuzz_CachedAccumulatorMatchesNaiveSlotsAndIndependentEconomicModel`, and `invariant_MiningPendingAndTpsCachesMatchEverySlot`.

Accepted consequence (finding M-01). Because incumbents retain their tenure rate while new tenures divide the *current* global rate by sixteen, aggregate issuance can exceed the current global rate after a halving. The excess persists until the legacy tenures turn over.

A HALVING DOES NOT REPRICE ANYONE



The shaded band is real: while pre-halving tenures survive, total issuance sits above the current global rate. Accepted deliberately, so that nobody can change the deal a miner already paid for.

HALVINGS APPLY TO NEWLY TAKEN SLOTS, NEVER TO SITTING MINERS. THE EXCESS IS TRANSITIONAL, BOUNDED BY TURNOVER, AND ACCEPTED AS THE PRICE OF NOT CHANGING A DEAL AFTER IT IS PAID FOR.

12.7 Next opening price

Formula F-6.

```
nextInitialPrice = clamp( ⌊p · m / 10^18⌋ , P_min , 2^192 − 1 )
```

A fill at `p = 0` yields `⌊0⌋ = 0`, which clamps up to `P_min`. Recovery from the floor is therefore geometric at rate `m` per fill, not immediate. Verified by `test_AFreeFillAtFullDecayRestartsAtTheConfiguredFloor` and `test_RecoveryFromTheFloorIsOnlyGeometric`.

12.8 Fixed slot topology

Mine constructs exactly sixteen empty slots and exposes no owner or capacity mutation. This permanent topology maps directly to a 4-by-4 market. Pending-emission and redemption-supply reads remain constant time regardless of how many slots are occupied.

16 PERMANENT SLOTS

fixed at construction · no owner · no capacity setter

held `1.00×`	held `1.00×`	**for sale** price falling	held `0.50×`
held `1.00×`	held `0.50×`	held `0.50×`	held `0.25×`
held `0.50×`	empty routes 100%	held `0.25×`	held `0.25×`
held `0.25×`	held `0.25×`	held `0.25×`	held `0.25×`

Each slot runs its own auction, falling to zero over 1 hour, and pays 80% to whoever it displaces.
A slot's issuance rate is fixed when it is taken and never moves again. The mixed rates above are incumbents who took their slots under different halvings.

THE MINING MARKET IS A FIXED BOARD OF SIXTEEN INDEPENDENT SLOTS. NOTHING CAN ADD A SEVENTEENTH, AND NO SLOT'S RATE CAN BE CHANGED BY ANYONE ONCE IT IS TAKEN.

PENDING EMISSION, ALL SIXTEEN SLOTS

Different start times, different locked rates, all accruing at once.


```
pendingEmission = stored + (now − updatedAt) × aggregateTps
```

The total is one multiplication no matter how many slots are occupied, which is why redemption can count unminted mining without touching the mine.

A handoff mints for the replaced slot only. The other fifteen are not read, checkpointed, or disturbed.

SIXTEEN INDEPENDENT TENURES, ONE CONSTANT-TIME TOTAL. REDEMPTION READS THE ACCUMULATOR; A HANDOFF SETTLES ONLY THE SLOT THAT CHANGED HANDS.

13. SignalGBX

`SignalGBX` is `ERC20, ERC20Votes, ReentrancyGuard, Ownable`. Name `"Signal GUM BALL 6900"`, symbol `"sGBX"`, 18 decimals.

13.1 Non-transferability

```
function _update(address from, address to, uint256 value) internal override(ERC20, ERC20Votes) {
    if (from != address(0) && to != address(0)) revert TransferDisabled();
    super._update(from, to, value);
}
```

Only mint (`from == 0`) and burn (`to == 0`) are permitted. `transfer` , `transferFrom` , self-transfer, and zero-value transfer all revert, with or without an allowance.

13.2 Collateralization

Identity I-3.

```
SignalGBX.totalSupply() ≤ GBX.balanceOf(SignalGBX)
```

with equality absent direct GBX donations. `_depositAndMint` transfers exactly `amount` GBX in and mints exactly `amount` sGBX; `_burnAndWithdraw` burns exactly `amount` and transfers exactly `amount` out. Both directions verify sender debit and receiver credit and revert `InexactUnderlyingTransfer` on mismatch. Direct GBX donations to the contract are stranded surplus that creates no receipt, signal, withdrawal entitlement, or voting power.

Verified by `testFuzz_SignalMoveWithdrawRoundTripIsLossless` , `invariant_SignalReceiptIsFullyCollateralized` , and `test_DirectDonationIsSurplusAndCreatesNoSignalVotesOrWithdrawalEntitlement` .

13.3 Mandatory signal-backing

ADR 0031 removed the separate `allocatedBalance` ledger, `_allocate` , `_deallocate` , and the `ISignalGBXAllocation` interface. Minting and burning are atomically coupled to the matching Resonance and paired-Bribe virtual-balance change, so **an idle receipt state is unreachable**.

Identity I-4 (mandatory signal-backing). Across live **and** killed Strategies, at every block:

```
SignalGBX.balanceOf(a)   = Σ_s Bribe(s).balanceOf(a)
SignalGBX.totalSupply()  = Σ_s Bribe(s).totalSupply()
GBX.balanceOf(SignalGBX) ≥ SignalGBX.totalSupply()
```

`Resonance.accountSignalWeight(a)` therefore returns `signalGBX.balanceOf(a)` directly rather than reading a second ledger. There is no reachable successful state in which a newly minted raw unit is idle, or a burned raw unit leaves signal behind.

Verified by `invariant_EveryReceiptUnitIsAssigned` , `invariant_SignalWeightNeverExceedsTheReceiptBalance` , `invariant_AccountWeightsSumToAllRecordedStrategyWeight` , and `test_RemovedIdleReceiptSelectorsAreAbsentFromRuntime` — which asserts the removed selectors are absent from deployed **runtime**, not merely from source.

`moveSignal` alters neither side of I-4: it changes which Bribe holds the balance, not the total.

13.4 Voting

`SignalGBX` inherits `ERC20Votes` on OpenZeppelin's default block-number clock (`CLOCK_MODE = mode=blocknumber`). Inside `_depositAndMint`:

```
if (delegates(account) == address(0)) _delegate(account, account);
```

so a first signal activates voting weight without a second transaction. An account with an existing non-zero delegate keeps it; an account that explicitly delegated to zero re-self-delegates on its next signal.

`SignalGBX` has **no ERC-2612 permit**. It inherits `EIP712` solely for ERC20Votes delegation signatures. This is deliberate: a permit authorizes spending, and sGBX cannot be spent.

Consequence of I-4 for a future integration. Because no sGBX can be idle, the ERC20Votes total supply is exactly the total signal committed across all Strategies. Any external system that uses `getPastTotalSupply` as a quorum denominator therefore measures against economically active weight only, with no idle receipts inflating it. That is a useful token property, not a quorum guarantee — the core defines no quorum (§15.2, §15.3).

13.5 No lock

There is no timestamp, epoch, cooldown, or lock state anywhere in `SignalGBX`. Signaling, moving, and withdrawing may occur in consecutive blocks or the same transaction. Combined with checkpoints that survive withdrawal (§15.3), this means an external system must not assume that recorded voting weight implies a currently held position. §35.3 and §38.2 analyze the consequence.

14. Signaling lifecycle

SIGNALING IS ONE STEP, NOT THREE

All of this happens in a single transaction, or none of it happens.

Deposit GBX
held by the receipt contract

→

Receive sGBX
same amount, non-transferable

→

Assign to a Strategy
named at the same moment

one transaction · withdrawing reverses all three

There is deliberately no state in between. You cannot hold sGBX that is not pointed at something, so every unit of signal that exists is a unit of signal doing work.

DEPOSIT, RECEIPT, AND COMMITMENT ARE ONE ATOMIC STEP. THE ABSENCE OF AN IN-BETWEEN STATE IS WHAT MAKES THE AMOUNT OF SIGNAL THAT EXISTS AND THE AMOUNT DOING WORK THE SAME QUANTITY.

14.1 Entry points

The complete user-facing surface is **four** functions on `SignalGBX`. Each takes scalar absolute amounts; there is no batch, array, percentage, or whole-account reset surface.

FUNCTION	COMPOSITION
`signal(strategy, amount)`	`_depositAndMint` → `Resonance.addSignalFor` → `Bribe.deposit`
`signalWithPermit(strategy, amount, …)`	`try permit` → `_depositAndMint` → `Resonance.addSignalFor` → `Bribe.deposit`
`moveSignal(from, to, amount)`	`Resonance.moveSignalFor` only — no GBX moves, no mint or burn, supply unchanged
`withdrawSignal(strategy, amount)`	`Resonance.removeSignalFor` → `Bribe.withdraw` → `_burnAndWithdraw`

Every failed sub-operation reverts the complete transition.

Removed by ADR 0031: `stake`, `unstake`, `stakeAndSignal`, `stakeAndSignalWithPermit`, `removeSignal` (leaving sGBX idle), and `removeSignalAndUnstake`. This is a breaking interface change; the repository has no production deployment, so no compatibility shim was introduced.

The permit variant wraps `IERC20Permit(gbx).permit(...)` in `try/catch` and discards failures. This is safe because the subsequent exact `transferFrom` is the authoritative authorization and custody check: a front-runner consuming the signature leaves the resulting allowance intact, and any other permit failure simply leaves the caller's pre-existing allowance to satisfy the transfer, or the whole call reverts. A failed or pre-consumed permit cannot create an unbacked receipt or a partial signal.

14.2 Sole-coordinator restriction

`Resonance.addSignalFor`, `removeSignalFor`, and `moveSignalFor` carry `onlySignalGBX`, reverting `UnauthorizedSignalSource` for any other caller. There is deliberately no second user-facing coordinator, and no direct-signaling path on Resonance.

14.3 Canonical state ownership

Each level of the signal ledger has exactly one owner; no value is duplicated across contracts.

LEVEL	CANONICAL STORAGE	READ ACCESSOR
Account aggregate	`SignalGBX.balanceOf(a)`	`Resonance.accountSignalWeight(a)`
Account × Strategy	`Bribe(s).balanceOf[a]`	`Resonance.accountSignals(a, s)`
Strategy total	`Bribe(s).totalSupply`	`Resonance.strategySignalWeight(s)`
Active total (live)	`Resonance.totalSignalWeight`	direct

The account aggregate is the sGBX balance itself, not a second ledger: ADR 0031 removed `allocatedBalance` precisely because it would always have been identical to `balanceOf` (§13.3).

14.4 Checkpoint-before-mutate ordering

This ordering is the protocol's defense against same-transaction revenue capture (finding **A-11**).

OPERATION	CHECKPOINT PERFORMED BEFORE ANY WEIGHT MUTATION
`addSignalFor`	`_updateReward(strategy)`
`removeSignalFor`	`_updateReward(strategy)`
`moveSignalFor`	`_updateReward(fromStrategy)` **and** `_updateReward(toStrategy)`
`killStrategy`	`updateReward(strategy)` modifier
`notifyRevenue`	`updateReward(address(0))` modifier — global index only
`distribute`	`updateReward(strategy)` modifier
`Strategy.buy`	`Resonance.distribute(address(this))` before reading inventory

Additionally, `Bribe.deposit` and `Bribe.withdraw` each call `_checkpointAll(account)` and `_fundAllPendingRewards()` before mutating `totalSupply` or `balanceOf`.

Consequence. In a single transaction, no stream time elapses between operations, so `rewardPerToken` cannot advance. A flash-signal therefore accrues exactly zero newly-notified revenue and zero newly-notified Bribe rewards. A signal held across real elapsed time legitimately earns that interval's flow — the protocol provides no epoch, cooldown, or anti-churn guarantee beyond this ordering.

Verified by `test_FlashSignalWeightCannotRedirectANewNotification`, `test_FlashSignalWeightCannotStealAccruedBribeRewards`, `test_NewStrategyWeightReceivesOnlyPostEntryRevenue`, and `test_StrategyAddedAfterAccrualCannotClaimHistoricRevenue` (finding A-11).

15. Protocol administration and external governance

ADR 0034 removed `ProtocolGovernor` and the protocol `TimelockController` from the core. This section specifies the administration surface that remains and states precisely which guarantees the core does **not** provide as a result.

15.1 The complete owner-gated surface

`Resonance` is the only owned core contract. Its owner-gated functions are:

FUNCTION	SOURCE	EFFECT	REVERSIBLE?
`addStrategy(IERC20, Config)`	`Resonance.sol`	Deploys Strategy, BribeRouter, and Bribe; registers the payment token	No (kill only)
`killStrategy(address)`	`Resonance.sol`	Permanently excludes a Strategy from new signal and future revenue	**No**
`addBribeReward(address, address)`	`Resonance.sol`	Appends a reward token to a Strategy's Bribe, within `MAX_REWARD_TOKENS`	**No**
`setBribeBps(uint256)`	`Resonance.sol`	Sets the global signaler share, bounded `[0, MAX_BRIBE_BPS]` (ADR 0036)	Yes, prospectively
`setResonanceRouter(address)`	`Resonance.sol`	Binds the sole ResonanceRouter	Single-use, then closed
`transferOwnership` / `renounceOwnership`	OpenZeppelin `Ownable`	Moves or destroys the owner role	Not by the protocol

`setResonanceRouter` reverts with `ResonanceRouterAlreadySet` after its first success, so it is a deployment binding rather than a continuing authority. The four continuing capabilities are therefore exactly `addStrategy`, `killStrategy`, `addBribeReward`, and `setBribeBps`.

Enforced constraints on those calls. These are Solidity checks, not procedural expectations:

CONSTRAINT	MECHANISM
The final live Strategy cannot be killed	`if (liveStrategyCount == 1) revert FinalLiveStrategy(strategy)`
The signaler share can never exceed 20%	`if (newBribeBps > MAX_BRIBE_BPS) revert BribeBpsAboveMaximum()`
A rate change cannot reclassify a settled amount	Snapshotted per payment, before any payment-token interaction
A Strategy cannot be killed twice	`isStrategyAlive` check, `StrategyAlreadyDead`
sGBX cannot be a payment token or Bribe reward token	`ForbiddenPaymentToken`, `ForbiddenRewardToken`
Payment and reward tokens must be deployed code	`code.length == 0` rejection
Bribe reward tokens are append-only and capped at eight	`Bribe.MAX_REWARD_TOKENS`, `RewardAlreadyAdded`
Strategy auction parameters are bounded at construction	`Strategy` constructor range checks (§21.1)

15.2 Authority the core does not implement

The core contains **no** Governor, Timelock, generic executor, multicall relay, or provider-specific governance adapter. It therefore makes none of the following guarantees, and no repository document should assert them:

ABSENT GUARANTEE	CONSEQUENCE
Selector-bounded proposal filtering	The owner calls the three functions directly; no calldata filter exists
Proposal threshold, quorum, or support	The core defines none; `liveStrategyCount` is the only counting rule
Voting period or voting delay	The core defines none
Post-approval execution delay	Owner calls take effect in the calling transaction
Permissionless execution after a delay	Not applicable; there is no queue
Cancellation, veto, or guardian	No such role exists in the core
Sole-proposer closure	Not applicable
Immutable governance parameters	Not applicable; the core has no governance parameters to fix

The owner address itself is the entire authority model at this commit. Nothing in `packages/contracts/src` constrains who or what that address is.

15.3 SignalGBX voting checkpoints

`SignalGBX` is `ERC20, ERC20Votes, ReentrancyGuard, Ownable`. It retains vote checkpoints deliberately, for a future external integration, and the core assigns them no semantics.

PROPERTY	VALUE
Interface	`IVotes` via OpenZeppelin `ERC20Votes`
Clock	Default block-number clock; no `clock()` or `CLOCK_MODE` override
Transferability	None — `_update` reverts `TransferDisabled` unless `from` or `to` is zero
Delegation	Standard; `_depositAndMint` self-delegates any account with no delegate
Approval permit	Not implemented on sGBX; the underlying GBX carries `ERC20Permit`
Supply relationship	Every sGBX unit is backed one-for-one by escrowed GBX and assigned to one Strategy (§14)

Because `_depositAndMint` self-delegates on first deposit, an account that has never delegated still accrues voting weight from its first signal without a second transaction. This removes the former undelegated-supply liveness concern **as a property of the token**; it does not constitute a quorum guarantee, because the core defines no quorum.

Checkpoints survive withdrawal. `withdrawSignal` burns sGBX and writes a new checkpoint, but historical checkpoints at earlier blocks are immutable by construction. An account may acquire GBX, signal it, allow a block to pass, withdraw, and retain its recorded weight at that past block. Whether that is exploitable depends entirely on whether the selected external system reads historical balances and how it spaces its snapshot from its voting window. This is finding **G-01**, retained as an integration property rather than a core defect.

15.4 Requirements on the future integration

ADR 0034 makes deployment conditional on a later ADR that pins and reviews at least:

1. provider, exact release, deployed bytecode, and proxy or upgrade model;
2. plugin set, permission graph, root and admin holders, and any emergency path;
3. direct compatibility with SignalGBX voting checkpoints and delegation, including snapshot-to-vote spacing (G-01);
4. proposal creation, quorum, support, voting duration, execution, batching, cancellation, and delay semantics; and
5. the exact `Resonance` owner address and the transaction evidence proving the handoff.

Until every item is selected, tested, and recorded, no deployment is authorized. Aragon is under consideration; no provider is part of the reviewed protocol graph, and this document must not be read as selecting one.

15.5 Security consequences

- Analysis of snapshot borrowing, quorum liveness, permission graphs, upgrade authority, and execution behavior does not apply to this repository and must be repeated against the selected system's exact release.
- A compromised or careless `Resonance` owner can add Strategies, kill any Strategy except the last live one, register reward tokens up to the eight-token cap, transfer ownership, or renounce it. `renounceOwnership` would permanently freeze the Strategy set at its current membership.
- Owner authority does not reach mining parameters, mint authority, Fund assets, liquidity custody, auction mechanics, or the sixteen-slot count. Those are immutable or held by ownerless contracts (§9). The one economic parameter it does reach, the signaler share, is bounded in code to `[0, MAX_BRIBE_BPS]` and applies prospectively only, so no owner can reclassify a settled amount or drive Fund's cumulative share below 80%.
- A production deployment that retains the temporary setup owner is a protocol with an ordinary admin key. Removing that owner is a release gate, not a recommendation (findings **M-03**, **G-01**, **G-03**).

16. Resonance

`Resonance` is `ReentrancyGuard, Ownable`. It is a Synthetix-shaped rewarder in which the "stakers" are Strategies and the "stake" is signal weight.

16.1 State

```
struct Reward {
    uint256 periodFinish;       // exclusive end of the active seven-day schedule
    uint256 remainderFinish;    // exclusive end of the one-extra-unit-per-second window
    uint256 rewardRate;         // base raw units per second
    uint256 lastUpdateTime;     // last timestamp folded into rewardPerTokenStored
    uint256 rewardPerTokenStored;// cumulative index, 1e36 scaled
}
```

The token-keyed ledger shape is retained from the Bribe lineage, but `Resonance` permanently registers exactly one reward token: USDG, pushed in the constructor. `rewardTokens` has length 1 forever; there is no function to append.

Constants: `DURATION = 7 days = 604_800`, `REWARD_PRECISION = 10^36`.

16.2 Live-weight semantics

`totalSignalWeight` is the sum of `Bribe(s).totalSupply()` over **live** Strategies only. It is:

- incremented by `amount` in `addSignalFor`;
- decremented by `amount` in `removeSignalFor` **only if the Strategy is alive**;
- incremented by `amount` in `moveSignalFor` **only if the source Strategy is dead** (re-entering the live denominator);
- decremented by the Strategy's entire recorded weight in `killStrategy`.

This asymmetry is precisely what prevents a killed Strategy's weight from being subtracted twice.

17. Six-decimal USDG reward mathematics

This section is the protocol's most precision-sensitive arithmetic, because the reward token carries six decimals while the weight denominator carries eighteen.

17.1 The decimal problem

Under the intended deployment, USDG has 6 decimals and sGBX has 18. A naive Synthetix index at `1e18` precision would compute, for `E` raw USDG emitted against total weight `W`:

```
Δindex = ⌊ E · 10^18 / W ⌋
```

With `W = 4600·10^18` (a modest 4,600 sGBX) and `E = 10^6` raw units (1.00 USDG), this yields `⌊10^6 · 10^18 / 4.6·10^21⌋ = ⌊217.4⌋ = 217`, and the per-Strategy recovery `⌊weight · 217 / 10^18⌋` reintroduces a second floor. The compounding of two floors at insufficient precision would round small allocations entirely to zero.

Resolution. `REWARD_PRECISION = 10^36`, chosen so the index carries `10^36 / 10^18 = 10^18` units of precision per unit of weight — restoring 18 significant digits of headroom against a 6-decimal reward.

17.2 Index formulas

Formula F-9 (cumulative index).

```
rewardPerToken() = rewardPerTokenStored                               if W = 0
                 = rewardPerTokenStored + mulDiv(E, 10^36, W)         otherwise

where W = totalSignalWeight
      E = emissionBetween(lastUpdateTime, min(now, periodFinish))
```

Symbols. `E` in raw USDG units; `W` in raw sGBX units (18 decimals); index in `1e36`-scaled USDG-per-raw-sGBX. *Rounding.* Floor. The residue `E·10^36 mod W` is **discarded, not carried** (§17.5). *Overflow.* `mulDiv` computes the 512-bit product `E · 10^36` before dividing. `E` is bounded by the scheduled amount; even `E = 2^96` leaves the product below `2^216`. No overflow is reachable.

Formula F-10 (Strategy entitlement).

```
earned(s) = account_Token_Rewards[s]
          + mulDiv( activeBalance(s), rewardPerToken() − paid[s], 10^36 )

where activeBalance(s) = isStrategyAlive[s] ? strategySignalWeight(s) : 0
```

Rounding. Floor. The residue `activeBalance · Δ mod 10^36` is **discarded, not carried** (§17.5). *Note.* A killed Strategy has `activeBalance = 0`, so `earned` returns exactly its stored pre-kill amount forever.

17.3 The raw emission schedule

Formula F-11 (schedule restart). On a qualifying notification of `reward` at time t_0 with remaining `left`:

```
S              = reward + left
rewardRate     = ⌊ S / 604800 ⌋
rateRemainder  = S mod 604800
periodFinish   = t₀ + 604800
remainderFinish= t₀ + rateRemainder
lastUpdateTime = t₀
```

Formula F-12 (emission between two timestamps). For `from < to`:

```
emissionBetween(from, to) = (to − from) · rewardRate
                          + ( from < remainderFinish
                              ? min(to, remainderFinish) − from
                              : 0 )
```

Lemma (schedule exactness). Emission over the full period equals `S` exactly:

```
emissionBetween(t0, t0 + 604800) = 604800 · rewardRate + (remainderFinish − t0)
                                 = 604800 · ⌊S/604800⌋ + (S mod 604800)
                                 = S    ∎
```

So **no raw USDG unit is lost to the rate division.** The remainder is emitted at one extra raw unit per second across the first `rateRemainder` seconds. This holds even for `S = 1`: `rewardRate = 0`, `rateRemainder = 1`, and the single raw unit is emitted during the first second. Verified by `test_OneRawUnitEmitsDuringTheFirstActiveSecond` and `test_RawRemainderIsFrontLoadedAndTheCompleteAmountIsScheduled`.

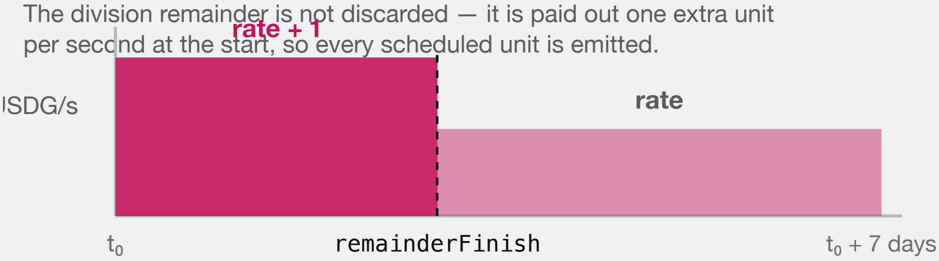
Formula F-13 (remaining).

```
left() = 0                                          if now ≥ periodFinish
       = emissionBetween(now, periodFinish)         otherwise

getRewardForDuration() = rewardRate · 604800 + (remainderFinish − (periodFinish − 604800))
```

ONE SEVEN-DAY SCHEDULE

The division remainder is not discarded — it is paid out one extra unit per second at the start, so every scheduled unit is emitted.



A one-raw-unit schedule still emits: the rate is zero and the single unit lands in second one.

REVENUE IS RELEASED OVER SEVEN DAYS AT A BASE RATE PLUS A FRONT-LOADED REMAINDER, SO THE SCHEDULE EMITS ITS EXACT TOTAL.

Worked example A (clean division). `S = 604_800_000_000` raw USDG (604,800.000000 USDG). `rewardRate = 1_000_000`, `rateRemainder = 0`. Emission is exactly 1 USDG per second for seven days.

Worked example B (with remainder). `S = 1_000_000` raw (1.00 USDG). `rewardRate = ⌊1_000_000/604_800⌋ = 1`; `rateRemainder = 1_000_000 − 604_800 = 395_200`. For the first 395,200 seconds emission is 2 raw units per second; thereafter 1 raw unit per second. Total `= 604_800·1 + 395_200 = 1_000_000`. Exact.

17.4 Allocation worked example

Three Strategies, weights `w_A = 1000·10^18`, `w_B = 3000·10^18`, `w_C = 600·10^18`, so `W = 4600·10^18`. One day elapses under Worked example A's schedule, emitting `E = 86_400 · 10^6 = 86_400_000_000` raw USDG.

Index advance: `Δ = mulDiv(86_400_000_000, 10^36, 4600·10^18) = ⌊8.64·10^46 / 4.6·10^21⌋ = 18_782_608_695_652_173_913_043_478` (units of `1e36`-scaled USDG per raw sGBX).

Per-Strategy recovery:

STRATEGY	WEIGHT (RAW SGBX)	`MULDIV(W, Δ, 10^36)`	USDG
A	`1000·10^18`	`18_782_608_695`	18,782.608695
B	`3000·10^18`	`56_347_826_086`	56,347.826086
C	`600·10^18`	`11_269_565_217`	11,269.565217
Sum		**`86_399_999_998`**	**86,399.999998**

Residue: 2 raw units (0.000002 USDG) unallocated. This is the compounded global-index and per-Strategy floor. It remains in Resonance permanently as surplus.

17.5 Accepted surplus — what Resonance deliberately does *not* conserve

Resonance has **three** sources of permanently unclassified USDG, and carries none of them:

1. **Global-index floor.** `E · 10^36 mod W` per checkpoint (§17.2, F-9).
2. **Per-Strategy floor.** `activeBalance · Δ mod 10^36` per Strategy per checkpoint (§17.2, F-10).
3. **Zero-active-weight emission.** `rewardPerToken()` returns early when `W = 0`, but `_updateReward` still advances `lastUpdateTime` to `lastTimeRewardApplicable()`. Stream time therefore elapses with **no** Strategy credited, and that interval's emission is permanently unclaimable.

Plus a fourth category that is never scheduled at all: **direct USDG transfers to Resonance**, since scheduling occurs only inside `notifyRevenue`.

This is a deliberate divergence from Bribe, which conserves sub-unit carry exactly (§23.4). ADR 0029 accepted the divergence (finding **A-02**) on the grounds that `1e36` precision makes individual floors economically negligible and that exact carry would require the additional state and boundary machinery Bribe carries.

> **No exact conservation identity and no lifetime dust bound is claimed for Resonance.** There is no synchronization, sweep, rescue, or later-allocation path. Checkpoint frequency and protocol lifetime determine the accumulation, and neither is bounded.

The strongest provable statement is an inequality (§30, Identity I-7).

18. Reward-period notification behavior

18.1 Router retention rule

`ResonanceRouter.route()` is permissionless and behaves as:

```
pending = USDG.balanceOf(router)
if pending = 0:             revert NoRevenue
minimum = Resonance.left(USDG)
if pending < minimum:       emit RevenueHeld(caller, pending, minimum); return 0
forward the ENTIRE pending balance; require balanceOf(router) = 0 afterwards
```

There is **no absolute minimum**. Because `left()` decays monotonically to zero at `periodFinish`, any retained balance eventually qualifies without further deposits. The rule prevents a sub-threshold Mine payment or fee harvest from reverting upstream while also preventing cheap stream-reset griefing.

18.2 Resonance acceptance rule

`notifyRevenue(reward)` is `onlyResonanceRouter`, runs `updateReward(address(0))` first, then:

```
if reward = 0:              revert ZeroAmount
remaining = left(USDG)
if reward < remaining:      revert RewardSmallerThanLeft
pull exactly `reward` from the router (exact-delta checked)
restart schedule with S = reward + remaining      (F-11)
```

18.3 Economic consequence of the restart rule

A restart moves `periodFinish` to `now + 7 days` and sets the rate to `⌊(reward + left)/604800⌋`. This can **raise or lower** the instantaneous rate:

- If `reward` is large relative to `left`, the rate rises.
- If `reward ≈ left` and substantial time has elapsed, the rate can fall, because the same remaining value is re-spread over a fresh seven days.

An actor wishing to force an early reset must supply at least `left`, so the manipulation is economically self-financing rather than free. The residual timing influence is intentional and accepted (§34.3).

Verified by `test_QualifyingTopUpCheckpointsAndRestartsWithRewardPlusLeft`, `test_TopUpBelowLeftRevertsAtomicallyAtResonance`, `test_SubThresholdRevenueWaitsUntilTheRouterBalanceQualifies`, and `test_AnOutsiderCannotExtendOrSlowTheLiveStream`.

18.4 Contrast with Bribe

The two streaming mechanisms behave **oppositely** on a top-up and must never be described interchangeably:

BEHAVIOR ON FUNDING DURING AN ACTIVE STREAM	`RESONANCE`	`BRIBE`
Minimum accepted amount	`≥ left`	any nonzero amount
Effect on the active schedule	checkpoints and **restarts** at `now`	**queues** behind it, undisturbed
Effect on `periodFinish`	moves to `now + 7 days`	unchanged
Sub-unit carry	discarded as surplus	conserved exactly, classified to Fund
Zero-supply behavior	time elapses, emission unclaimable	stream **pauses**, time preserved

19. Signal checkpoint ordering

The ordering discipline of §14.4 has a precise formal statement.

Property P-1 (no same-transaction capture). Let `τ` be a transaction containing a weight mutation at internal step `j` and any revenue-bearing operation at step `k > j`. Because every mutation and every notification calls `_updateReward` before mutating, and because `rewardPerToken` advances only as a function of `lastTimeRewardApplicable() - lastUpdateTime`, which is identically zero within one `block.timestamp`, the index cannot advance between steps `j` and `k`. Therefore weight introduced at step `j` earns exactly zero from any emission recognized at step `k`.

Property P-2 (no retroactive redirection). Symmetrically, weight *removed* at step `j` retains its full entitlement to all emission checkpointed before step `j`, because `_updateReward(strategy)` settles `account_Token_Rewards[strategy]` into storage before the weight changes.

Corollary. The pair (P-1, P-2) means signal weight earns exactly the emission that elapses while it is allocated — no more, no less, up to the floors of §17.5.

What P-1 does not provide. It bounds capture within a transaction, not across blocks. A signaler who allocates and holds across real elapsed time earns that interval's flow legitimately, and may unallocate immediately afterwards. There is deliberately no epoch, cooldown, minimum duration, or anti-churn mechanism. Rapid allocation movement and wallet-splitting are permitted by design.

19.1 Strategy purchase ordering

`Strategy.buy` calls `ICoreResonance(resonance).distribute(address(this))` **before** reading `revenueToken.balanceOf(address(this))`. Consequently:

- the buyer receives every USDG unit released to the Strategy through the execution timestamp, not merely its stale visible balance; and
- combined with P-1, a same-transaction signal-then-buy sequence can acquire only inventory that predated the routed payment.

This is the direct remediation of finding **A-11**.

20. Strategy registration and lifecycle

20.1 Registration

`Resonance.addStrategy(paymentToken, config)` is `onlyOwner nonReentrant` and performs, in order:

1. Reject `paymentToken` that is zero or code-less; reject `paymentToken == signalGBX` (`ForbiddenPaymentToken`).
2. `bribeFactory.createBribe()` → new `Bribe` bound to this Resonance, reading `fund` from it.
3. `bribe.addRewardToken(paymentToken)` — the payment token occupies reward slot 1 of 8 automatically.
4. `strategyFactory.createStrategy(usdg, paymentToken, fund, bribe, config)` → deploys `Strategy` **and** `BribeRouter` together.
5. Record `isStrategy`, `isStrategyAlive`, `bribeFor`, `bribeRouterFor`, `paymentTokenFor`.

6. Set `account_Token_RewardPerTokenPaid[strategy][usdg] = rewardPerTokenStored`.

Step 6 is what prevents a newly registered Strategy from claiming historical revenue: it starts at the current index, so its first `earned` computation sees `Δ = 0`. Verified by `test_StrategyAddedAfterAccrualCannotClaimHistoricRevenue`.

The rejection of sGBX as a payment token (finding **E-03**) exists because sGBX transfers are permanently disabled: a Strategy priced in sGBX could never be filled, and the reward slot it consumed in the append-only Bribe registry could never be reclaimed.

Both factories reject any caller other than their permanently bound Resonance (`NotResonance`), so there is no public Strategy or Bribe creation path.

20.2 Death

`killStrategy(strategy)` is `onlyOwner nonReentrant updateReward(strategy)`:

```
require isStrategy[strategy] ∧ isStrategyAlive[strategy]
require liveStrategyCount ≠ 1                          -- else revert FinalLiveStrategy
isStrategyAlive[strategy] ← false
--liveStrategyCount
totalSignalWeight ← totalSignalWeight − strategySignalWeight(strategy)
```

The `updateReward` modifier runs first, so the Strategy's accrued whole USDG units are settled into `account_Token_Rewards` and remain permanently claimable via `distribute`. Thereafter `earned` uses `activeBalance = 0`, so no further accrual occurs.

Death is **irreversible**. There is no revive function. Since ADR 0031, the **final** live Strategy cannot be killed at all: `liveStrategyCount == 1` reverts `FinalLiveStrategy`, so at least one valid signal destination always exists (§29.3).

20.3 Post-death signal semantics

OPERATION ON A KILLED STRATEGY `S`	PERMITTED?	EFFECT ON `TOTALSIGNALWEIGHT`
`addSignalFor(a, s, x)`	No	reverts `StrategyAlreadyDead`
`removeSignalFor(a, s, x)`	**Yes**	**unchanged** — weight was already excluded at kill
`moveSignalFor(a, s, → live, x)`	**Yes**	**increased by** `x` — re-enters the live denominator
`moveSignalFor(a, live, → s, x)`	No	reverts `StrategyAlreadyDead`

The asymmetry is essential: subtracting on `removeSignalFor` would double-subtract weight already removed by `killStrategy`. Verified by `test_DeadStrategySignalCanExitWithoutSubtractingActiveSupplyTwice` and `test_MoveFromKilledStrategyReentersLiveWeightExactlyOnce`.

20.4 The closed-pool consequence (finding BR-1)

`Bribe` has no kill state and no awareness that its Strategy died. It becomes a **closed pool**:

- `deposit` is unreachable, because the only caller is `Resonance.addSignalFor`, which rejects dead Strategies.

- Incumbent signalers may remain indefinitely, earn and claim independently funded rewards, and exit incrementally.
- When `totalSupply` reaches zero, `withdraw` pauses every stream. A paused stream resumes only via `deposit` — which can never occur again.
- Queued rewards likewise require a `deposit` to start.
- `notifyRewardAmount` remains callable, so tokens can still be added to a pool with zero possible claimants.

The abandoned amount is not bounded to dust. It may include a complete unvested stream plus any later notification. There is deliberately no retirement state, refund, rescue, sweep, or Fund reclassification. Accepted by ADR 0028; evidenced by `test_KnownRisk_DeadStrategyBribeCanPauseAndQueueRewardsForever`.

Interfaces **must** identify dead Strategies, warn the final signaler that exiting abandons remaining rewards, and must not imply that a notification to a dead zero-supply Bribe is recoverable.

21. Reverse Dutch acquisition auctions

21.1 Immutable configuration

PARAMETER	SYMBOL	BOUND
`epochDuration`	`D_s`	`3600 ≤ D_s ≤ 31_536_000` (1 hour … 365 days)
`priceMultiplier`	`m_s`	`1.1·10^18 ≤ m_s ≤ 3·10^18`
`minimumPrice`	`p_min`	`10^6 ≤ p_min ≤ 2^192−1`
`initialPrice`	`p_0`	`p_min ≤ p_0 ≤ 2^192−1`

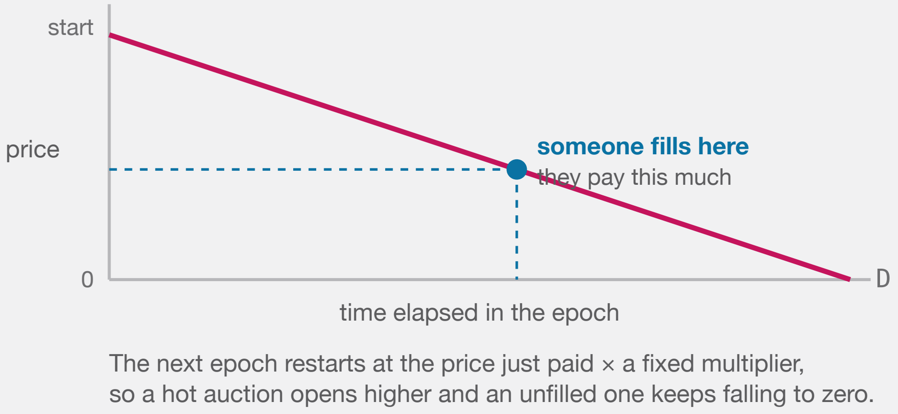
`PRICE_SCALE = 10^18`, `ABSOLUTE_MINIMUM_PRICE = 10^6`, `ABSOLUTE_MAXIMUM_PRICE = 2^192−1`.

21.2 Price function

Formula F-14. For `e = now − epochStartedAt`:

```
currentPrice(e) = initialPrice − ⌊ initialPrice · e / D_s ⌋     for e < D_s
                = 0                                               for e ≥ D_s
```

Units. Raw units of the **payment token**, whose decimals are that token's own. *Rounding.* Floor on the subtracted term, so the quoted price is at or above the ideal line. Monotonically non-increasing within an epoch. *Overflow.* `mulDiv`; `initialPrice < 2^192` bounds the intermediate.



BOTH THE MINING SLOTS AND THE STRATEGY AUCTIONS USE THE SAME SHAPE: THE ASKING PRICE FALLS IN A STRAIGHT LINE TO ZERO, AND WHOEVER THINKS IT IS CHEAP ENOUGH FILLS IT.

Important distinction. `p_min` is a floor on the **next epoch's starting price**, not on the fill price. Fills below `p_min`, including at exactly zero after full decay, are normal and expected.

Worked example. `D_s = 86_400` (24 h), `p_0 = 4·10^8` raw (4.00000000 WBTC at 8 decimals). At `e = 47_520` (55% elapsed): `price = 4·10^8 − ⌊4·10^8 · 47_520 / 86_400⌋ = 4·10^8 − 2.2·10^8 = 1.8·10^8` = 1.8 WBTC.

21.3 Fill

```
buy(revenueReceiver, expectedEpochId, deadline, maximumPayment):
  require revenueReceiver ≠ 0
  require now ≤ deadline                              else DeadlinePassed
  require expectedEpochId = epochId                   else EpochIdMismatch
  Resonance.distribute(this)                          -- pull released USDG first
  revenueAmount ← revenueToken.balanceOf(this)
  require revenueAmount ≠ 0                           else EmptyRevenue
  paymentAmount ← currentPrice()
  require paymentAmount ≤ maximumPayment              else MaximumPaymentExceeded
  if paymentAmount ≠ 0:
      pull exactly paymentAmount (exact-delta checked)
      _settlePayment(paymentAmount)
  transfer exactly revenueAmount to revenueReceiver (exact-delta checked)
  initialPrice ← nextInitialPrice(paymentAmount)
  epochStartedAt ← now
  epochId ← epochId + 1
```

Buyer protections are `expectedEpochId` (defeats a competing fill changing the price mid-flight), `deadline`, and `maximumPayment`.

Formula F-15 (next starting price).

```
nextInitialPrice = clamp( ⌊ paymentAmount · m_s / 10^18 ⌋ , p_min , 2^192 − 1 )
```

A zero-price fill yields `p_min`. Recovery is geometric at `m_s` per fill. Verified by `test_OneLateFillCollapsesTheAuctionToItsFloor` and `test_RecoveryFromTheFloorIsOnlyGeometric`.

21.4 Receiver semantics

`revenueReceiver` is buyer-chosen and unvalidated beyond non-zero. Consequences, all tested:

RECEIVER	OUTCOME
Ordinary address	Normal
`Fund`	Settles exactly; USDG becomes Fund backing (`test_RevenueReceiverEqualToFundSettlesExactly`)
`Resonance`	Creates **unscheduled surplus** — the USDG is not re-notified
The `Strategy` itself	Fails atomically (exact-delta check cannot be satisfied)

22. Auction-payment settlement

22.1 Settlement path

`Strategy._settlePayment(amount)`:

```
router ← Resonance.bribeRouterFor(this)
require router ≠ 0
paymentToken.forceApprove(router, amount)
BribeRouter(router).routePayment(amount)
if paymentToken.allowance(this, router) ≠ 0: paymentToken.forceApprove(router, 0)
```

The conditional zero-approval (finding **E-04**) supports tokens that revert on redundant zero approvals.

22.2 The bounded classification (ADR 0036)

`BribeRouter` holds no setter and no share of its own. It reads one global rate from `Resonance` and derives Fund's share as the remainder:

```
uint256 public constant BPS = 10_000;              // BribeRouter

uint256 public constant DEFAULT_BRIBE_BPS = 1_000; // Resonance: 10% at deployment
uint256 public constant MAX_BRIBE_BPS     = 2_000; // Resonance: 20% ceiling
uint256 public bribeBps = DEFAULT_BRIBE_BPS;       // Resonance: onlyOwner setBribeBps
```

The rate is snapshotted before any token interaction, so a hostile payment-token callback cannot alter the split of the fill it is part of:

```
uint256 appliedBribeBps = ICoreResonance(resonance).bribeBps();
if (appliedBribeBps > BPS) revert BribeBpsAboveBasis(appliedBribeBps);
```

Because `MAX_BRIBE_BPS = 2_000`, Fund's cumulative share over the complete weighted payment history is **at least 80%** for any sequence of rates the owner can select. A rate change reaches neither a recorded liability, a notified reward, nor a prior Fund balance.

`routePayment(amount)` is callable **only** by its immutable `strategy`:

```
appliedBribeBps      ← Resonance.bribeBps()          (snapshotted before any token call)
require appliedBribeBps ≤ BPS
pull exactly `amount` from strategy (exact-delta checked)

bribeAmount          ← ⌊amount · appliedBribeBps / BPS⌋
accumulatedRemainder ← splitRemainder + mulmod(amount, appliedBribeBps, BPS)
bribeAmount          ← bribeAmount + ⌊accumulatedRemainder / BPS⌋
splitRemainder       ← accumulatedRemainder mod BPS
fundAmount           ← amount − bribeAmount

accountedPaymentBalance += amount
fundPaymentLiability    += fundAmount
bribePaymentLiability   += bribeAmount
```

This **supersedes ADR 0021's 100%-to-Fund rule.** The split applies to the **acquired payment asset**; USDG is unaffected, since Resonance still transfers 100% of a Strategy's earned USDG to that Strategy (§16).

Overflow. `Math.mulDiv` computes the quotient at 512-bit intermediate width and `mulmod` computes the modulus without overflowing, so no intermediate product can wrap.

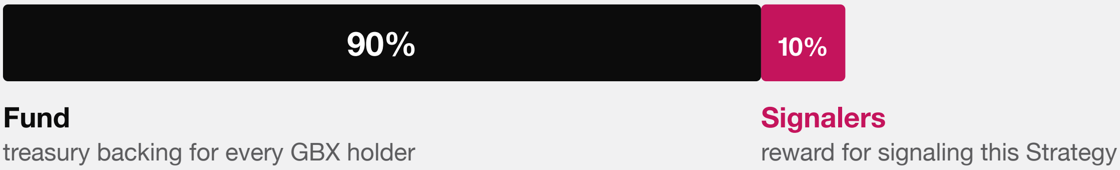
22.3 Cumulative exactness

Formula F-21 (frequency-independent classification). `splitRemainder` carries the sub-unit Bribe entitlement in basis-point numerator units and is always `< BPS`. For payments `a_i` classified at their snapshotted rates `r_i`, **regardless of how the total was partitioned into calls**:

```
cumulative Bribe classification = ⌊(Σ aᵢ·rᵢ) / BPS⌋
cumulative Fund classification  = (Σ aᵢ) − ⌊(Σ aᵢ·rᵢ) / BPS⌋
splitRemainder                  = (Σ aᵢ·rᵢ) mod BPS
```

At a constant rate this reduces to the single-rate form. Since every `rᵢ ≤ MAX_BRIBE_BPS`, the cumulative Fund share is bounded below by 80% of the weighted total for any admissible rate history.

EVERY ACQUIRED PAYMENT



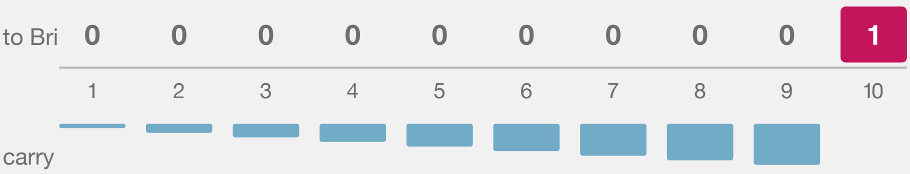
Fixed in code. No setter, no governance parameter, no caller-chosen destination.

EVERY ASSET THE PROTOCOL ACQUIRES IS DIVIDED BY AN IMMUTABLE RULE THE MOMENT IT ARRIVES.

Why the carry is load-bearing. Naive per-payment flooring would give the Bribe `⌊1 · 1000 / 10000⌋ = 0` on every one-raw-unit payment at the default rate, so an adversary filling in dust could starve the reward share permanently. This was internal finding **SR-002** (High).

TEN SEPARATE ONE-UNIT PAYMENTS

Naive flooring would give the Bribe zero every time, forever.
The carried remainder makes the tenth payment settle the debt exactly.



Cumulative result: Fund 9, Bribe 1, remainder 0 — identical to one lump payment of ten.

WHY THE SPLIT CARRIES ITS REMAINDER: WITHOUT IT, AN ADVERSARY PAYING IN DUST WOULD STARVE THE REWARD SHARE PERMANENTLY.

Worked example (SR-002's minimal trace). Ten separate one-raw-unit payments:

CALL	`AMOUNT`	`ACCUMULATEDREMAINDER` BEFORE ÷	`BRIBEAMOUNT`	`FUNDAMOUNT`	`SPLITREMAINDER` AFTER
1	1	1,000	0	1	1,000
2	1	2,000	0	1	2,000
…	…	…	…	…	…
9	1	9,000	0	1	9,000
10	1	10,000	**1**	**0**	**0**

Cumulative state is exactly **Fund 9, Bribe 1, remainder 0** — matching `⌊10 · 1000 / 10000⌋ = 1`. Verified by `test_TenOneUnitPaymentsClassifyExactlyNineToFundAndOneToBribe`, `test_TenOneUnitPaymentsDoNotStarveTheBribe`, and `testFuzz_ClassificationIsFrequencyIndependent`.

`splitRemainder` is a fractional entitlement in numerator units — never a withdrawable token balance, never a caller-controlled destination.

22.4 Isolated deferred settlement legs

Both legs are permissionless, and each clears its own state **before** its external interaction so a failure atomically restores only that leg:

```
payFundPayment():
    amount ← fundPaymentLiability ; if 0 return
    fundPaymentLiability ← 0 ; accountedPaymentBalance -= amount
    transfer exactly `amount` to fund (exact-delta checked)

notifyBribeReward():
    amount ← bribePaymentLiability ; if 0 return
    bribePaymentLiability ← 0 ; accountedPaymentBalance -= amount
    forceApprove(bribe, amount) ; bribe.notifyRewardAmount(paymentToken, amount)
    clear residual allowance if nonzero ; verify exact debit and credit
```

A failure or incompatibility on one leg cannot destroy, consume, or block permissionless retry of the other. Deferral is what prevents a frozen or hostile Fund, Bribe, or payment token from blocking an auction fill.

The Bribe leg always has a valid registered reward token, because `addStrategy` registers the Strategy's payment token as reward slot 1 of 8 at creation (§20.1).

Verified by `test_PayingFundIsPermissionlessAndClearsTheLiability`, `test_NotifyingBribeIsPermissionlessExactAndClearsOnlyItsLeg`, `test_AFailureOnEitherSettlementLegDoesNotBlockOrCorruptTheOther`, `test_BribeNotificationRejectsReentrancyAndStillVerifiesExactDeltas`, and `test_BribeNotificationClearsAStickyResidualAllowance`.

22.5 Settlement identities

Identity I-5 (BribeRouter exactness). At every block:

```
accountedPaymentBalance = fundPaymentLiability + bribePaymentLiability
paymentToken.balanceOf(router) ≥ accountedPaymentBalance
paymentSurplus() = paymentToken.balanceOf(router) − accountedPaymentBalance     (direct donations)
splitRemainder < BPS
```

Proof sketch. All three counters start at zero. `routePayment` adds `amount` to the first and `fundAmount + bribeAmount = amount` across the other two. Each settlement function subtracts the identical amount from `accountedPaymentBalance` and its own liability. ∎

Identity I-6 (payment conservation). For any completed fill at nonzero price `p`:

```
p = increment to fundPaymentLiability + increment to bribePaymentLiability
```

with **zero** routed to Resonance, to a fee recipient, or to any caller-selected destination. Direct donations change neither liability nor `splitRemainder`.

Verified by `test_CompletePaymentIsClassifiedNinetyTenEvenWithLiveSignalWeight`, `testFuzz_ClassificationIsFrequencyIndependent`, `test_RoutePaymentIsStrategyOnly`, and `invariant_BribeRouterAccountingIdentitiesAreExact`.

22.6 GBX-denominated Strategies

A Strategy whose payment token is GBX is handled identically — no special case exists. The 90% share travels `buyer → Strategy → BribeRouter → Fund` and is **supply-neutral until explicitly burned**; `Fund.burnGBX(amount)` is permissionless and burns from Fund's own balance. The 10% share is streamed to that Strategy's signalers as a GBX reward and is **not** burned.

Operational consequence. Fund-held GBX is included in `gbx.totalSupply()`, which is the redemption denominator (§25.2). Unburned Fund GBX therefore inflates the denominator and reduces every redeemer's payout. A redeemer should settle and burn pending Fund GBX before quoting or executing a redemption.

23. Bribe reward accounting

`Bribe` is the protocol's exact-carry accounting subsystem, and is materially more intricate than Resonance because it conserves what Resonance discards.

Constants: `REWARD_DURATION = 7 days`, `REWARD_PRECISION = 10^18`, `MAX_REWARD_TOKENS = 8`.

23.1 State inventory

MAPPING	MEANING
`scheduledRewards[t]`	Whole units remaining in the active stream
`queuedRewards[t]`	Whole units waiting for the stream to finish or supply to appear
`pendingRewardScaled[t]`	Emitted precision not yet large enough for an index increment
`indexedRewardScaled[t]`	Precision indexed globally but not yet folded into accounts
`rewards[a][t]`	Whole-unit user liability, payable only to `a`
`userRewardRemainder[a][t]`	Sub-unit user carry retained across checkpoints
`accruedRewardLiability[t]`	Σ over accounts of `rewards[a][t]`
`fundRewardLiability[t]`	Whole-unit liability irrevocably owed to Fund
`fundRewardRemainder[t]`	Sub-unit Fund carry awaiting a whole unit
`accountedRewardBalance[t]`	Notified minus paid out (user + Fund)
`lifetimeRewardNotified[t]`	Monotonic cumulative raw units ever admitted for `t` (ADR 0035)

23.2 Stream mathematics

Structurally identical to F-11/F-12 at `REWARD_DURATION`, with one addition: a `pauseStarted` field.

```
_startStream(t, amount, startedAt):
    rewardRate      ← ⌊amount / 604800⌋
    remainder       ← amount mod 604800
    periodFinish    ← startedAt + 604800
    remainderFinish ← startedAt + remainder
    lastUpdateTime  ← startedAt
    pauseStarted    ← 0
    scheduledRewards[t] ← amount
```

Formula F-16 (accrual).

```
_accrueUntil(t, ts):
    emitted ← emissionBetween(lastUpdateTime, ts)
    lastUpdateTime ← ts
    scheduledRewards[t]     −= emitted
    pendingRewardScaled[t]  += emitted · 10^18
```

Formula F-17 (indexing).

```
_indexPendingReward(t):
    if supply = 0: return
    δ ← ⌊ pendingRewardScaled[t] / supply ⌋
    if δ = 0: return
    indexed ← δ · supply
    pendingRewardScaled[t] −= indexed
    indexedRewardScaled[t] += indexed
    rewardPerTokenStored   += δ
```

The residue `pendingRewardScaled mod supply` is **retained**, not discarded — this is the essential difference from Resonance.

Formula F-18 (account checkpoint).

```
newlyIndexed ← balanceOf(a) · (rewardPerTokenStored − paid[a])
indexedRewardScaled[t] −= newlyIndexed
soleCarry ← (balanceOf(a) ≠ 0 ∧ balanceOf(a) = totalSupply) ? pendingRewardScaled[t] : 0
             (and pendingRewardScaled[t] ← 0 in that case)
accrued ← userRewardRemainder[a][t] + newlyIndexed + soleCarry
whole   ← ⌊accrued / 10^18⌋
userRewardRemainder[a][t] ← accrued mod 10^18
rewards[a][t] += whole ; accruedRewardLiability[t] += whole
```

The sole-signaler branch exists because when one account holds the entire supply, the global residue is unambiguously theirs. `earned()` mirrors it by adding `globalScaled mod supply` in the same condition.

23.3 Queueing and pausing

Queueing. `notifyRewardAmount` starts a stream immediately **only if** `totalSupply ≠ 0` **and** `periodFinish = 0`. Otherwise the amount joins `queuedRewards`. A live stream is therefore never reset, shrunk, or extended by a top-up — this defeats repeated-tiny-top-up griefing.

`_checkpointToken` advances through at most the current stream and **one** queued successor per call, bounding work.

Pausing. When `totalSupply` reaches zero in `withdraw`, `_pauseStream` records `pauseStarted = now` for every token. When supply leaves zero in `deposit`, `_resumeAllStreams` adds `pausedDuration = now − pauseStarted` to `periodFinish`, `remainderFinish`, and `lastUpdateTime`, preserving the schedule's remaining shape exactly. While paused, `_checkpointToken` and `_previewEmission` return early and `lastTimeRewardApplicable` returns `pauseStarted`.

Verified by `test_ZeroSignalWeightPausesAndExtendsTheStreamWithoutRetroactiveAccrual` and `testFuzz_RepeatedZeroSupplyPausesPreserveEveryEarlyRemainderUnit`.

23.4 Carry classification to Fund

Before **every** supply change, `deposit` and `withdraw` call `_fundAllPendingRewards()`, which for each token moves the entire `pendingRewardScaled` into Fund classification:

```
_accrueFundScaled(t, scaled):
    s ← fundRewardRemainder[t] + scaled
    whole ← ⌊s / 10^18⌋
    fundRewardRemainder[t] ← s mod 10^18
    fundRewardLiability[t] += whole
```

Additionally, when an account's balance reaches zero in `withdraw`, its `userRewardRemainder` for every token is moved to Fund and deleted.

Rationale (finding A-09, ADR 0027). Carry accumulated under one denominator cannot be fairly indexed under a different one. Leaving it in `pendingRewardScaled` would let a **later entrant** receive value emitted before their entry, or let **remaining signalers** absorb an exited account's precision. Routing it to Fund makes it protocol backing instead — value that no individual can capture.

Verified by `test_NewSignalerCannotReceivePreEntryRewardCarry`, `test_RemainingSignalerCannotReceivePreExitRewardCarry`, and `test_FullExitCannotReallocateUserRewardRemainder`.

23.5 Bounded reward registry

`MAX_REWARD_TOKENS = 8`, append-only, `onlyResonance`. The cap is what makes every mandatory loop — `_checkpointAll`, `_fundAllPendingRewards`, the exit carry loop, the pause loop, the resume loop — constant-bounded (finding **A-08**). Gas is measured by `test_MaximumRewardTokenGasStaysFarBelowABlock` and `test_RewardTokenGasSlopeIsRecordedAndBounded`.

23.6 Lifetime notification cap (ADR 0035)

Each `(Bribe, token)` pair carries a monotonic counter of every raw unit ever admitted through `notifyReward-Amount`, whether the notifier is a `BribeRouter` settling the automatic 10% share or an independent funder. The immutable ceiling is:

Formula F-24.

```
P    = REWARD_PRECISION = 1e18
MAX_LIFETIME_REWARD_AMOUNT = ⌊(2²⁵⁶ − 1) / P⌋
```

A notification is rejected with `RewardLifetimeCapExceeded(token, notified, requested, maximum)` when

```
amount > MAX_LIFETIME_REWARD_AMOUNT − lifetimeRewardNotified[t]
```

The check runs **before** any checkpoint or token interaction, so a rejection leaves the caller's balance, every schedule, and every liability untouched.

Why the counter is monotonic. Claims, Fund classification, Fund payment, stream completion, Strategy death, and a return to zero signal supply all reduce `accountedRewardBalance[t]`, but none of them reduces the cumulative reward-per-signal index. The pre-existing balance-scale guard (`RewardScaleOverflow`) therefore reopened capacity that the index had already consumed. A token with an extremely large raw-unit supply could fill the index near `uint256` maximum, reclaim its reward, and notify again; the next checkpoint would overflow. Because every signal deposit, move, and withdrawal checkpoints all registered tokens, that overflow would strand escrowed GBX. This was finding **BR-2**.

Safety argument. One admitted raw unit contributes at most `P` scaled units to the global index, and the smallest possible virtual supply is one raw signal unit, which assigns the whole scaled amount. With lifetime notifications `N`:

```
rewardPerTokenStored ≤ N · P     and     N ≤ ⌊(2²⁵⁶ − 1) / P⌋
```

so the stored and previewed index remain representable. A one-unit virtual supply attains the bound, making this the largest history-independent limit that is safe under arbitrary supply changes.

Consequences. At 18 decimals the cap is approximately `1.158 × 10⁴¹` whole tokens and constrains no conventional asset; the limit is deliberately measured in raw units, so it can bind an unusually high-decimal token. Reaching it blocks only new notifications for that one token in that one Bribe — existing rewards, claims, signal moves, and withdrawals continue (§32, L-9). If an automatic Strategy-payment reward is rejected, its `BribeRouter` preserves the unpaid Bribe liability and the independently settleable Fund liability, so no value is consumed or redirected (§22.4). Direct token donations never enter the index and never consume the cap. The balance-scale guard remains as defense in depth, and no unchecked wrapping, epoch reset, retirement withdrawal, or rescue path is introduced.

23.7 Claim isolation

Three claim entry points: all-tokens, single-token, and caller-selected list (with duplicate and registration validation performed **before** any token interaction). All pay the entitled `account`, never `msg.sender`. Selective claiming is what lets a signaler omit a broken reward token without losing access to the others.

23.8 Exit liveness

`Bribe.withdraw` performs **only** accounting: checkpoints, carry classification, balance decrements, and stream pauses. It contains no `transfer`, `transferFrom`, or `safeTransfer`. Therefore signal withdrawal can never fail because a reward, payment, or revenue token is frozen — design goal G4. Verified by `invariant_EveryActorCanFullyWithdrawSignals`.

24. Fund custody

`Fund` is `ReentrancyGuard` only. It is **not** `Ownable`, has no roles, and its only storage is the immutable `gbx`.

24.1 Value exits

Exactly two, both permissionless:

FUNCTION	EFFECT
`burnGBX(amount)`	Burns `amount` of Fund's **own** GBX balance
`redeem(...)`	Burns caller's GBX, transfers floored pro-rata basket

There is no sweep, rescue, recovery, migration, or administrative withdrawal of any kind. Assets omitted by every redeemer remain in Fund indefinitely, backing the residual supply.

24.2 Registry-free design

Fund maintains **no asset list**. Any ERC-20 transferred to it becomes redeemable backing without review, registration, or governance action.

Therefore: presence in Fund does **not** indicate protocol endorsement. Official protocol membership is represented solely by Strategies registered in `Resonance`. Interfaces must label unsolicited Fund balances separately, support manual asset-address entry, and warn that omissions are permanently forfeited.

25. GBX redemption

25.1 Interface

```
function redeem(uint256 gbxAmount, address receiver, address[] calldata tokens) external
nonReentrant
```

Validation: `gbxAmount ≠ 0`; `receiver ∉ {0, address(this)}`; `tokens.length ≠ 0`; every entry unique and not GBX or zero (§25.4).

25.2 Algorithm

```
1. require gbx.minterLocked() ∧ mine.code.length > 0 ∧ IMine(mine).gbx() = gbx
2. supplyBeforeBurn ← IMine(mine).effectiveTotalSupply() -- minted plus all pending mining issuance
3. require supplyBeforeBurn ≠ 0 ∧ gbxAmount ≤ supplyBeforeBurn
4. for each token i:
       _markToken(i)                                  -- transient duplicate guard
       balancesBefore[i] ← IERC20(i).balanceOf(Fund)
       payouts[i]        ← mulDiv(balancesBefore[i], gbxAmount, supplyBeforeBurn)
5. transferFrom(msg.sender → Fund, gbxAmount) ; gbx.burn(gbxAmount)
6. for each token i:
       require IERC20(i).balanceOf(Fund) ≥ balancesBefore[i]     -- pre-transfer guard
       if payouts[i] ≠ 0: transfer exactly payouts[i] to receiver (exact-delta checked)
       _clearToken(i)
7. for each token i:
       require IERC20(i).balanceOf(Fund) ≥ balancesBefore[i] − payouts[i]   -- final basket guard
```

Formula F-19 (payout).

```
payout_i = ⌊ balanceOf_i(Fund) · gbxAmount / supplyBeforeBurn ⌋
```

Units. Raw units of token `i`. *Rounding.* Floor — the residue stays in Fund for the residual supply, so rounding always favors remaining holders. *Overflow.* `mulDiv` at 512-bit intermediate precision.

Every payout uses the **same** `supplyBeforeBurn`, so a multi-asset basket is internally consistent. The entire operation is atomic: any revert unwinds the burn and all transfers.

25.3 Why effective supply includes pending mining

Mining accrual is lazy (§12.6): a miner's earned GBX is unminted until its slot changes hands. Reading only `totalSupply()` would exclude already-earned issuance, and a redeemer would receive a share computed against an artificially small denominator — diluting miners in favor of redeemers. `effectiveTotalSupply()` eliminates the timing advantage in constant time and does not mutate Mine. This is the remediation of finding **A-10**.

25.4 Duplicate detection via EIP-1153

```
slot = keccak256(abi.encode(REDEMPTION_NAMESPACE, token))
_markToken:  reject token ∈ {0, gbx};  if tload(slot) ≠ 0 revert DuplicateToken;  tstore(slot, 1)
_clearToken: tstore(slot, 0)
```

This gives `O(n)` duplicate detection with **no** permanent storage writes, no sorting requirement, no asset registry, no token IDs, and no monotonic nonce. Marks are cleared after each successful payout so that a second redemption later in the same transaction is fully independent.

Deployment prerequisite. The target chain must support EIP-1153 (Cancun). This is a hard requirement recorded in `docs/TRUST_ASSUMPTIONS.md`.

25.5 The shared-ledger guard

Steps 6 and 7 defeat an attack in which two distinct addresses front the same underlying balance ledger. Without the final pass, a redeemer could nominate both facades and extract the same backing twice. The final pass requires each selected address to retain at least `balancesBefore[i] - payouts[i]`, catching asymmetric aliases where only the later transfer mutates the earlier address's reported balance. This is the remediation of finding **E-01**, evidenced by `test_RedeemRejectsDifferentAddressesThatDebitOneSharedLedger` and `test_RedeemFinalPassRejectsAnAsymmetricAliasSideEffect`.

REDEEMING

Burn your GBX → **Name the assets** you want → **Your slice**

For each asset you name:

`floor( Fund's balance × your GBX ÷ total GBX supply )`

The supply is snapshotted after every miner is credited, so nobody is diluted by unminted rewards. Assets you leave out are forfeited — permanently, to everyone else.

It is one atomic step. If any transfer fails, the burn is undone too.

REDEMPTION IN FULL. NO QUEUE, NO GATE, NO DISCRETION — ONLY ARITHMETIC AGAINST A SNAPSHOT.

25.6 Worked redemption example

State. Fund holds `50 WBTC` (`5·10^9` raw at 8 decimals) and `400 ETH` (`4·10^20` raw at 18 decimals). `gbx.totalSupply() = 10^8 · 10^18`. Miners have accrued `10^6 · 10^18` unminted GBX. Redeemer burns `gbxAmount = 2.5·10^5 · 10^18` (250,000 GBX).

Step 2. `effectiveTotalSupply()` returns `1.01·10^26` without minting or changing a slot.

Step 4.

```
WBTC: ⌊ 5·10^9 · 2.5·10^23 / 1.01·10^26 ⌋ = ⌊ 12_376_237.62 ⌋ = 12_376_237 raw = 0.12376237 WBTC
ETH : ⌊ 4·10^20 · 2.5·10^23 / 1.01·10^26 ⌋ = 990_099_009_900_990_099 raw ≈ 0.990099009900990099 ETH
```

Step 5. Supply falls to `1.0075·10^26` (100,750,000 GBX).

Counterfactual. Without step 2, `supplyBeforeBurn = 10^26` and the WBTC payout would be `12_500_000` raw (0.125 WBTC). The checkpoint costs this redeemer `123_763` raw WBTC — exactly the dilution attributable to the miners' already-earned claim, and therefore correct.

This reproduces the mechanism modelled in `packages/simulations/fixtures/economic-scenarios.json`, whose analogous scenario yields `495_049_504_950` raw USDG with the checkpoint versus `500_000_000_000` without.

26. LiquidityPosition

`LiquidityPosition` is `IERC721Receiver, ReentrancyGuard`. It is **ownerless** and has **no withdrawal function of any kind**.

26.1 Admission

`onERC721Received` accepts exactly one NFT, requiring simultaneously:

```
msg.sender = positionManager            -- only the canonical PositionManager may deliver
¬positionRecorded                       -- one-time
from       = positionDepositor          -- precommitted depositor
tokenId    = expectedPositionTokenId    -- precommitted ID
ownerOf(tokenId) = address(this)
keccak256(abi.encode(receivedPoolKey)) = poolKeyHash
tickLower  = expectedTickLower  ∧  tickUpper = expectedTickUpper
getPositionLiquidity(tokenId) ≠ 0
```

The constructor additionally requires the pool key's currencies to be exactly `{GBX, USDG}` in address order, the hooks address to be zero (`NonzeroHook` — the pool must be hookless), `tickLower < tickUpper`, and reciprocal token identity on both destinations (`router.usdg() == usdg`, `fund.gbx() == gbx`).

Once admitted, the NFT can never leave, by any caller, through any mechanism. Admission checks run once, on receipt, and are the only defense against a misconfigured position.

26.2 Fee harvesting

Formula F-20 (harvest).

```
principal ← getPositionLiquidity(tokenId)
modifyLiquidities([DECREASE_LIQUIDITY(tokenId, 0, 0, 0, ""), CLOSE_CURRENCY(c0),
CLOSE_CURRENCY(c1)], now)
require getPositionLiquidity(tokenId) = principal          else PrincipalLiquidityChanged
usdgRouted ← USDG.balanceOf(this) ; if ≠ 0: transfer to router (exact) ; router.route()
gbxBurned  ← GBX.balanceOf(this)  ; if ≠ 0: transfer to fund (exact) ; fund.burnGBX(gbxBurned)
```

A zero-liquidity `DECREASE_LIQUIDITY` is Uniswap v4 PositionManager's canonical fee-collection path; the two `CLOSE_CURRENCY` actions take the fee credits without touching principal. The post-condition check is an explicit guard that principal is genuinely unchanged.

`harvestFees` is **permissionless with no caller bounty**. Routing and burning are atomic with collection: any failure reverts the whole harvest, restoring the position's fee accounting. Direct GBX or USDG donations to the contract are swept to the same fixed destinations on the next harvest.

26.3 Genesis position properties

The position is intended to begin as a **one-sided, out-of-range GBX-only** position funded entirely by the 20,000,000 GBX genesis allocation, so that bootstrap requires no matching stablecoin capital and the position sells into demand as price rises.

> **This is a deployment property, not a contract guarantee.** `LiquidityPosition` enforces the pool identity, the tick range, hooklessness, and nonzero liquidity. It does **not** verify one-sidedness, that the range is out of market, or the deposited amount. An incorrect genesis price or range strands the position permanently and cannot be corrected.

27. Ownership lifecycle and the enforcement boundary

§15 specifies the owner-gated surface. This section specifies **who holds that owner role over time**, and draws the line between what Solidity enforces and what a deployment must prove.

27.1 Ownership at each stage

STAGE	`RESONANCE.OWNER()`	CAPABILITY HELD
Construction	`initialOwner` constructor argument	All owner-gated calls, including `setResonanceRouter`
Bootstrap	Temporary deployment setup owner	Binds the Router; creates reviewed initial Strategies
After handoff (**required**)	Exact reviewed external executor	`addStrategy`, `killStrategy`, `addBribeReward`
Optional terminal state	`address(0)` after `renounceOwnership`	None; Strategy membership is frozen permanently

`SignalGBX`, `StrategyFactory`, and `BribeFactory` each retain a nominal `Ownable` owner after their one-time `setResonance` binding is consumed, but no remaining function is gated on it (§9.1). `Mine`, `Fund`, `LiquidityPosition`, `Strategy`, and `BribeRouter` are not `Ownable` at all. `Bribe` gates `addRewardToken` on the immutable `resonance` address rather than on an owner.

27.2 What Solidity enforces, and what it does not

This distinction is the single most important qualification in this document.

Enforced by the contracts, unconditionally and without reference to any deployment procedure:

- `GBX.setMinter` succeeds at most once and permanently sets `minterLocked` (§11).
- `Mine` has no owner and no capacity-changing function; `SLOT_COUNT` is `constant` (§12).
- `Resonance.setResonanceRouter`, `SignalGBX.setResonance`, `StrategyFactory.setResonance`, and `BribeFactory.setResonance` each revert after their first success.
- One-time bindings validate reciprocal identity — a Mine must report the same GBX, a Router the same Resonance and USDG, a Resonance the same SignalGBX and factory pair (§28.1).
- `killStrategy` reverts on the final live Strategy.
- `Fund`, `LiquidityPosition`, `Strategy`, and `BribeRouter` expose no owner, sweep, rescue, or migration path.

Not enforced by any contract in this repository, and therefore a deployment obligation that must be proven by signed evidence rather than asserted:

- That the `Resonance` owner after handoff is the intended external governance executor rather than an EOA, a compromised address, or a lookalike contract.
- That the temporary setup owner retains no authority afterward.

- That constructor arguments — mining rate, halving amount, tail rate, price multiplier, minimum initial price, pool key, tick range, precommitted position token ID — carry the exact reviewed values.
- That the deployed dependencies are the canonical USDG and Uniswap v4 contracts on the target chain.

Reciprocal identity checks reject a *crossed* protocol graph. They cannot distinguish a malicious lookalike that returns the expected identities, and the protocol has no upgrade, successor, or migration authority with which to repair a wrong value. This is finding **M-03** — an open High release gate — and the residue of **E-02**.

Where any repository document states an ownership or role condition as an "invariant", it is correctly read as a **deployment obligation**. This was recorded as discrepancy D-5 (§43).

27.3 Handoff sequence and evidence

Ownership handoff is step 10 of `docs/DEPLOYMENT.md`, and it is gated on step 9: deployment stops unless a later ADR has selected and reviewed the external governance integration. The sequence is:

```
1. Setup owner binds ResonanceRouter (single-use).
2. Setup owner creates every reviewed bootstrap Strategy and registers reviewed Bribe reward tokens.
3. A later ADR pins the external governance provider, release, bytecode, permission graph, and vot-
ing semantics.
4. Setup owner calls transferOwnership(exact reviewed executor). No intermediate custodian.
5. Verify Resonance.owner(), the handoff receipt, and that the coordinator retains no authority.
```

Bootstrap Strategies are created **before** handoff deliberately: the initial membership is part of the reviewed deployment rather than the first act of an unreviewed governance system. The consequence is that the setup owner's key is a full-authority key until step 4 completes, and a deployment interrupted between steps 2 and 4 leaves a protocol with a live admin key. There is no contract-level timeout, escrow, or forced handoff.

28. Deployment and immutable bindings

28.1 Binding table

BINDING	GUARD	RECIPROCAL CHECK	REUSABLE?
`GBX.setMinter(Mine)`	`msg.sender = minter`, `¬minterLocked`	`IMine(m).gbx() = GBX`	No
`SignalGBX.setResonance(R)`	`onlyOwner`, unset	`R.signalGBX() = SignalGBX`	No
`StrategyFactory.setResonance`	`onlyOwner`, unset	`R.strategyFactory() = this`	No
`BribeFactory.setResonance`	`onlyOwner`, unset	`R.bribeFactory() = this`	No
`Resonance.setResonanceRouter`	`onlyOwner`, unset	`Router.resonance() = this` **and** `Router.usdg() = usdg`	No
`Mine` constructor	—	`Router.usdg() = usdg`	n/a
`LiquidityPosition` ctor	—	`Router.usdg() = usdg`, `Fund.gbx() = gbx`	n/a
`Bribe` constructor	—	reads `fund` from the bound `Resonance`	n/a

Every reciprocal read is `try/catch` guarded and reverts on failure or on a mismatched identity.

`SignalGBX` additionally refuses **all** staking and signaling until its Resonance binding completes (`ResonanceNotSet`), so no user can deposit into a partially-wired graph.

28.2 What reciprocal checks do and do not prove

They prove **consistency**: contract A and contract B agree they refer to each other, and to the same USDG or GBX. They cannot prove **honesty**: a malicious lookalike that returns the expected identities passes every check.

This materially reduces accidental cross-wiring but does not close finding **M-03**, which remains an open High release gate requiring exact runtime code hashes, constructor arguments, transaction receipts, and a signed manifest.

28.3 Deployment order (summary)

The intended sequence from `docs/DEPLOYMENT.md`: deploy GBX with a temporary coordinator as minter → deploy Fund, SignalGBX, both factories → deploy Resonance with a temporary setup owner, bind it into SignalGBX and both factories, deploy ResonanceRouter and bind it → deploy Mine and verify its identities → `GBX.setMinter(Mine)` **irreversibly** → create every reviewed bootstrap Strategy while the setup owner still controls Resonance → initialize the pool and create the precommitted position → deploy LiquidityPosition and safe-transfer the NFT → **stop** unless a later ADR has selected and reviewed the external governance integration → transfer Resonance ownership directly to the exact reviewed external executor and prove the coordinator retains no authority → reconcile all runtime bytecode, arguments, bindings, ownership, and custody (§27.3).

28.4 The irreversibility budget

Every one of the following is permanent and unrepairable once executed:

ACTION	FAILURE MODE IF WRONG
`GBX.setMinter`	Wrong or malicious issuer forever; no second minter possible
Any `setResonance` / router binding	Permanently crossed graph
Mine constructor economics	Wrong emission curve forever
`Resonance` ownership handoff	Wrong or hostile administrator for the three capabilities
Pool key, tick range, NFT ID	Genesis liquidity stranded permanently
NFT safe-transfer to LiquidityPosition	Position locked forever regardless of correctness
Retaining the temporary setup owner	A live admin key in a protocol that claims to have none
Bootstrap Strategy set	Unwanted Strategies exist forever (killable, not removable)

A failed setup must be abandoned entirely before use. There is no repair authority.

29. State machines

29.1 Mining slot

FROM	TRANSITION	TO	EFFECTS
Empty	`mine` at price `p`	Occupied	100% of `p` routes; `tps ← ⌊globalTps/16⌋`; `epochId++`
Occupied	`mine` at price `p > 0`	Occupied	settle this slot; `⌊0.8p⌋` → claim, `⌈0.2p⌉` → router; assign new `tps`
Occupied	`mine` at price `p = 0`	Occupied	settle this slot; **no token movement**; assign new `tps`; `epochId++`
Occupied	`Fund.redeem`	Occupied	no slot mutation; effective supply includes the slot's pending emission

A slot never returns to Empty. The permanent slot count is sixteen.

29.2 Signal position (account × Strategy)

FROM	TRANSITION	TO	GUARD
None	`signal` / `signalWithPermit`	Signalling	Strategy live; caller holds `x` GBX and allowance
Signalling	`signal`	Signalling (+x)	Strategy live; caller holds `x` GBX and allowance
Signalling	`withdrawSignal`	Signalling (−x)	`x ≤ Bribe.balanceOf(a)`; burns sGBX, returns GBX
Signalling	`moveSignal(s → s')`	Signalling on `s'`	`s'` live; `s ≠ s'`; `x ≤ Bribe(s).balanceOf(a)`; supply unchanged
Signalling on dead	`withdrawSignal`	Signalling (−x)	permitted; active weight unchanged
Signalling on dead	`moveSignal(dead → live)`	Signalling	re-enters the live denominator exactly once
Signalling on dead	`signal`	—	**reverts** `StrategyAlreadyDead`

Because of Identity I-4 there is no `None → Idle` transition and no idle state at all: a position either exists on some Strategy or the sGBX does not exist.

29.3 Strategy

FROM	TRANSITION	TO	NOTES
—	`addStrategy`	Live	Deploys Strategy + BribeRouter + Bribe; `++liveStrategyCount`; index initialized to current
Live	`killStrategy` when `liveStrategyCount > 1`	Dead	Checkpoints first; weight excluded; `--liveStrategyCount`; **irreversible**
Live	`killStrategy` when `liveStrategyCount = 1`	Live	**reverts** `FinalLiveStrategy` — a replacement must be added first
Dead	—	Dead	No revive transition exists

The final-live-Strategy guard (ADR 0031, superseding ADR 0029's permission to kill it) guarantees a valid signal destination always exists — necessary now that signaling is the only way to hold sGBX. The owner replaces the last Strategy by calling `addStrategy(replacement)` before `killStrategy(previous)`; whether those two calls can be atomically batched is a property of the external governance system, not of the core. Verified by `test_Killing-TheFinalLiveStrategyRevertsAfterBootstrap`.

29.4 Strategy auction epoch

FROM	TRANSITION	TO	EFFECTS
Epoch n	`buy` at price `p > 0`	Epoch n+1	distribute → pull `p` → settle → pay out USDG → reprice
Epoch n	`buy` at price `p = 0`	Epoch n+1	distribute → **no payment** → pay out USDG → `p_min` restart
Epoch n	`buy` with zero USDG	Epoch n	**reverts** `EmptyRevenue`
Epoch n	`buy` with stale `epochId`	Epoch n	**reverts** `EpochIdMismatch`

29.5 Bribe reward stream (per token)

FROM	TRANSITION	TO	EFFECTS
Inactive	`notify` with `totalSupply > 0`	Active	`_startStream` at `now`
Inactive	`notify` with `totalSupply = 0`	Inactive	amount → `queuedRewards`
Active	`notify` (any supply)	Active	amount → `queuedRewards`; **stream undisturbed**
Active	`totalSupply → 0` via `withdraw`	Paused	`pauseStarted ← now`; pending carry → Fund
Paused	`totalSupply > 0` via `deposit`	Active	all boundaries shifted by `pausedDuration`
Active	`now ≥ periodFinish`, queue empty	Inactive	`_clearFinishedStream`; index preserved
Active	`now ≥ periodFinish`, queue nonempty	Active	successor started at old `periodFinish`
Paused	Strategy dead, no signaler can return	**Terminal**	**rewards permanently unclaimable** (BR-1)

29.6 Fund liability (BribeRouter and Bribe)

FROM	TRANSITION	TO	NOTES
Zero	auction fill / carry accrual	Outstanding	Liability recorded; destination immutable
Outstanding	`payFundPayment` succeeds	Zero	Exact transfer verified
Outstanding	`payFundPayment` reverts	Outstanding	Atomically restored; permanently retryable

29.7 `Resonance` ownership

FROM	TRANSITION	TO	NOTES
Constructor owner	Deployment bootstrap (§27.3)	Setup owner	Binds Router; creates reviewed bootstrap Strategies
Setup owner	`transferOwnership(executor)`	External executor	Required before any user funds; step 10 of deployment
Any owner	`transferOwnership(other)`	Other owner	Unconstrained by the core
Any owner	`renounceOwnership()`	`address(0)`	**Irreversible**; Strategy membership frozen permanently

The core imposes no delay, approval, confirmation, or two-step acceptance on any of these transitions. There is no state in which an ownership change is pending and observable before it takes effect.

30. Accounting identities

Only identities the implementation can actually prove are asserted. Where exact conservation does not hold, an inequality is stated instead and the residue is named.

I-1 — GBX supply. `totalSupply = lifetimeMinted − lifetimeBurned`. *Exact.* (§11.3)

I-2 — Mine solvency. `USDG.balanceOf(Mine) ≥ totalClaimable`, equality absent donations. *Exact modulo donations.* (§12.4)

I-3 — sGBX collateralization. `sGBX.totalSupply ≤ GBX.balanceOf(SignalGBX)`, equality absent donations. *Exact modulo donations.* (§13.2)

I-4 — Mandatory signal-backing. `∀a: sGBX.balanceOf(a) = Σ_s Bribe(s).balanceOf(a)`, and `sGBX.totalSupply() = Σ_s Bribe(s).totalSupply()`, across live and killed Strategies. *Exact.* (§13.3)

I-5 — BribeRouter exactness. `accountedPaymentBalance = fundPaymentLiability + bribePaymentLiability`, and `splitRemainder < BPS`. *Exact.* (§22.5)

I-6 — Signal ledger consistency.

```
∀a:  Σ_s Bribe(s).balanceOf(a)              = SignalGBX.balanceOf(a)
∀s:  Σ_a Bribe(s).balanceOf(a)              = Bribe(s).totalSupply
     Σ_{s live} Bribe(s).totalSupply        = Resonance.totalSignalWeight
```

Exact. Note the third line ranges over **live** Strategies only; a killed Strategy retains its recorded balances while contributing zero to the active total. Verified by `invariant_AccountWeightsSumToAllRecordedStrategyWeight`, `invariant_BribeBalancesMirrorAccountSignals`, `invariant_BribeSupplyMirrorsStrategyWeight`, `invariant_StrategyWeightsSumToTheGlobalTotal`, and `invariant_DeadStrategiesAreExcludedFromActiveWeight`.

I-7 — Resonance solvency (inequality only).

```
USDG.balanceOf(Resonance) = left(USDG) + Σ_s earned(s, USDG) + surplus,   surplus ≥ 0
```

where `surplus` comprises global-index floors, per-Strategy floors, zero-active-weight emission, and direct donations.

> **This is deliberately not an equality with a bounded residue.** No exact conservation and no lifetime dust bound is claimed for Resonance (§17.5). Verified as an inequality by `invariant_ResonanceIsSolventAgainstClaimableRevenue`, `invariant_ResonanceScheduledAndEarnedRevenueIsSolvent`, and `testFuzz_AccruedAndScheduledRevenueNeverExceedsTheHeldBalance`.

I-8 — Bribe conservation (exact). For each registered token `t`, every accounted unit is represented in exactly one place:

```
accountedRewardBalance[t] = scheduledRewards[t]
                          + queuedRewards[t]
                          + accruedRewardLiability[t]
                          + fundRewardLiability[t]
                          + ⌊(pendingRewardScaled[t] + indexedRewardScaled[t]
                              + Σ_a userRewardRemainder[a][t] + fundRewardRemainder[t]) / 10^18⌋
```

and `IERC20(t).balanceOf(Bribe) ≥ accountedRewardBalance[t]`, the difference being direct donations exposed by `rewardSurplus(t)`. *Exact.* Verified by `invariant_BribeAccountingIdentitiesAreExact`, `invariant_BribesAreSolventAgainstAccruedRewards`, `invariant_ScheduledRewardsNeverExceedHeldBalance`, and `testFuzz_BribeIsAlwaysSolventAgainstAccruedRewards`.

I-9 — Fund redemption conservation. For each selected token `i`:

```
balanceAfter_i = balanceBefore_i − ⌊balanceBefore_i · gbxAmount / supplyBeforeBurn⌋
supplyAfter    = supplyBeforeBurn − gbxAmount
```

and therefore backing per remaining GBX is non-decreasing:

```
balanceAfter_i / supplyAfter  ≥  balanceBefore_i / supplyBeforeBurn
```

Exact, with the inequality strict whenever the floor discards a residue. Verified by `testFuzz_PayoutIsExactlyTheFlooredProRataShare`, `testFuzz_BackingPerGBXNeverDecreasesOnRedemption`, `testFuzz_SequentialRedemptionsStaySolvent`, and `invariant_FundNeverOwesMoreThanItHolds`.

I-10 — Effective supply. `Mine.effectiveTotalSupply() = GBX.totalSupply() + Mine.pendingEmission()` before any checkpoint. *Exact.* Verified by `invariant_EffectiveSupplyIncludesEveryPendingEmission`.

I-11 — USDG conservation across the routing path. Every raw unit leaving `Mine` as routed revenue arrives at `ResonanceRouter`, and every unit forwarded from the Router arrives at Resonance, with exact-delta checks on both legs and `RevenueRetained` reverting any shortfall. Verified by `invariant_USDGIsConserved`, `testFuzz_RoutingConservesEveryUnit`, `testFuzz_RoutingIsExactlyConservative`, and `invariant_RevenueRouterRetentionIsFullyVisible`.

30.1 Identities deliberately *not* asserted

TEMPTING CLAIM	WHY IT IS NOT ASSERTED
Resonance USDG is exactly conserved	Three floor/discard sources; see I-7 and §17.5
Resonance dust is bounded over the protocol lifetime	Depends on unbounded checkpoint frequency and lifetime
Every Bribe reward is eventually claimable	False after BR-1 abandonment (§20.4)
Fund backing per GBX is non-decreasing under all operations	Only proven for redemption; unsolicited transfers and burns move it in either direction
Aggregate GBX issuance ≤ current global rate	False while pre-halving tenure rates remain locked (M-01, §12.6)
The `Resonance` owner is the intended administrator	Procedural, not code-enforced (§27.2)

31. Security invariants

ID	INVARIANT	EVIDENCE
S-1	Only the permanently bound Mine can mint GBX, and only after `minterLocked`	`test_OnlyPermanentlyBoundMineCanMint`
S-2	`slot.tps` is never rewritten during a tenure	`test_HalvingUsesEconomicAccrualAndNeverRepricesAnIncumbent`
S-3	Mine always has exactly 16 slots and no capacity mutation	`test_LaunchesWithSixteenEmptySlotsAndPermanentMiningAuthority`
S-4	sGBX is non-transferable in every path, including self and zero-value	`test_TransfersRemainPermanentlyDisabled`
S-5	Only SignalGBX may mutate signal state on Resonance	`test_OnlySignalGBXCanMutateAnotherAccountsSignal`
S-6	Only Resonance may mutate Bribe virtual balances or append reward tokens	`test_VirtualBalanceMutationIsResonanceOnly`
S-7	Only the bound Resonance may deploy through either factory	`test_FactoriesAreResonanceOnly`
S-8	Only the immutable Strategy may call `BribeRouter.routePayment`	`test_RoutePaymentIsStrategyOnly`
S-9	Only the bound Router may call `Resonance.notifyRevenue`	`test_NotifyRevenueIsRouterOnlyAndRejectsZero`
S-10	A same-transaction signal cannot capture newly notified revenue or Bribe rewards	`test_FlashSignalWeightCannotRedirectANewNotification`, `test_FlashSignalWeightCannotStealAccruedBribeRewards`
S-11	Bribe carry cannot cross a signal-supply boundary to a later entrant	`test_NewSignalerCannotReceivePreEntryRewardCarry`
S-12	An exiting account's remainder cannot be reallocated to remaining signalers	`test_FullExitCannotReallocateUserRewardRemainder`

ID	INVARIANT	EVIDENCE
S-13	Every value transfer verifies exact sender debit and receiver credit	the full fee-on-transfer rejection family (§36)
S-14	Redemption rejects GBX, zero, and duplicates in any position	`test_RedeemRejectsDuplicatesInAnyPosition`
S-15	A redemption basket cannot double-consume one shared backing ledger	`test_RedeemRejectsDifferentAddressesThatDebitOneSharedLedger`
S-16	Redemption is atomic: any failure reverts the burn and all transfers	`test_ASelectedFailingTransferRollsBackTheEntireRedemption`
S-17	Redemption includes all pending mining in a constant-time denominator	`test_RedemptionUsesEffectiveSupplyWithoutSettlingAnyMiner`
S-18	Reentrancy cannot double-claim a reward or double-fill an auction	`test_ReentrantRewardPayoutCannotDoubleClaim`, `test_AHostilePaymentTokenCannotReenterTheSameStrategy`
S-19	Harvesting never changes principal liquidity	`testFuzz_HarvestIsExactAndPrincipalIsFixed`
S-20	The canonical NFT can never leave LiquidityPosition	`test_TheCanonicalNFTCanNeverLeaveOnceAdmitted`
S-21	Each continuing administration call is owner-gated	`test_AddStrategyIsOwnerOnlyAndCreatesTheCompleteGraph`, `test_KillStrategyIsOwnerOnlyPermanentAndBlocksNewSignal`, `test_AddBribeRewardIsOwnerOnlyAndDelegatesToThePairedBribe`
S-22	Cumulative reward notifications per token cannot exhaust the reward index	`test_LifetimeRewardCapAcceptsTheExactLimitAndRejectsTheFirstExcessUnit`, `test_LifetimeRewardCapStillBlocksAfterTheMaximumWasClaimed`
S-23	Fund has no administrative surface	`test_FundHasNoAdministrativeSurfaceLeft`
S-24	Redemption and GBX burning are the only ways assets leave Fund	`test_RedemptionIsTheOnlyWayAssetsCanEverLeaveFund`
S-25	Reward-token registry is capped at 8 and append-only	`test_RewardTokenCountIsPermanentlyCappedAtEight`

32. Liveness properties

ID	PROPERTY	DEPENDS ON
L-1	An account can always withdraw signal it holds	GBX transferability only
L-2	An account can always remove signal, including from a dead Strategy	**Nothing** — pure accounting (§23.7)
L-3	Redemption cannot be paused, gated, or blocked by any party	The redeemer's own token selection
L-4	A redeemer can route around a broken asset by omitting it	Caller-selected basket
L-5	A signaler can claim one reward token while another is frozen	Selective claim
L-6	A Fund liability blocked by a hostile token remains observable and retryable	Token eventually functioning
L-7	Auction fills are never blocked by a frozen Fund	Deferred settlement (§22.2)
L-8	Sub-threshold revenue eventually enters the stream without action	`left()` decaying to zero (§18.1)
L-9	Signal exit stays available after a Bribe reaches its lifetime reward cap	`test_KilledStrategyExitRemainsLiveAfterRewardLifetimeCapIsConsumed`
L-10	Every mandatory loop is bounded	`MAX_REWARD_TOKENS = 8` ; Fund does not loop Mine slots

32.1 Liveness properties that do not hold

NON-PROPERTY	REASON
Rewards in a dead Strategy's Bribe are always claimable	BR-1 terminal state (§20.4)
Resonance surplus is always recoverable	No recovery path exists (§17.5)
Uniswap fees are always harvested	No caller bounty; harvesting is voluntary
Mining accrual is always current	Lazy; requires a voluntary checkpoint
Administration is always available	Depends entirely on the unselected external owner (§15, G-03)
An administration call can be observed before it lands	The core provides no delay, queue, or pending state (§29.7)

33. Precision and rounding analysis

33.1 Rounding inventory

#	SITE	FORMULA	DIRECTION	RESIDUE DESTINATION	BOUNDED?
1	Mine price decay	`⌊P·e/D⌋` subtracted	Favors protocol	none (price is quoted)	≤1 unit
2	Mine payment split	`⌊p·0.8⌋`	Favors protocol	routed as revenue	≤1 unit
3	Mine next price	`⌊p·m/10^18⌋`	Favors payer	none	≤1 unit

#	SITE	FORMULA	DIRECTION	RESIDUE DESTINATION	BOUNDED?
4	New tenure rate	`⌊globalTps/16⌋`	Reduces issuance	**unissued forever**	<16/s
5	Halving rate	`u_0 >> k`	Reduces issuance	unissued	≤1/s
6	Resonance schedule rate	`⌊S/604800⌋` + front-loaded remainder	**exact**	none — fully emitted	**0**
7	Resonance global index	`⌊E·10^36/W⌋`	Reduces payout	**Resonance surplus**	**No**
8	Resonance per-Strategy	`⌊w·Δ/10^36⌋`	Reduces payout	**Resonance surplus**	**No**
9	Bribe schedule rate	`⌊A/604800⌋` + front-loaded remainder	**exact**	none	**0**
10	Bribe global index	`⌊pendingScaled/supply⌋`	Deferred	**retained in `pendingRewardScaled`**	**0**
11	Bribe account settlement	`⌊accrued/10^18⌋`	Deferred	**retained in `userRewardRemainder`**	**0**
12	Bribe boundary classification	`⌊scaled/10^18⌋`	Deferred	`fundRewardRemainder` → **Fund**	**0**
13	Fund redemption payout	`⌊bal·g/S⌋`	Favors remaining holders	stays in Fund	≤1 unit/token

Rows 6, 9–13 are exact or fully carried. Rows 7 and 8 are the protocol's only unbounded value leakage.

33.2 Why 1e36 and not 1e18

With `E` raw USDG (6 decimals) and `W` raw sGBX (18 decimals), the relative truncation error of the global index is approximately `1/(E · P / W)` where `P` is the precision constant. Taking a representative `W = 10^22` (10,000 sGBX) and one second of a 1 USDG/second stream (`E = 10^6`):

PRECISION `P`	`⌊E·P/W⌋`	RELATIVE ERROR OF ONE CHECKPOINT
`10^18`	`⌊10^2⌋ = 100`	~1%
`10^36`	`⌊10^20⌋ = 10^20`	~10^{-20}

At `1e18`, per-checkpoint truncation of order 1% would be economically material and would compound with checkpoint frequency. At `1e36` it is negligible per checkpoint. Bribe can safely use `1e18` because its reward tokens are not constrained to six decimals *and*, more importantly, because it carries its residue rather than discarding it.

33.3 The decimal-asymmetry hazard

> **Six-decimal USDG is a deployment assumption, not a code constant.** No contract calls `usdg.decimals()` or validates it. The `1e36` calibration, every worked example in this document, and every economic figure in the simulation fixtures assume 6 decimals. Binding a USDG with different decimals would leave the contracts functional but invalidate the calibration and all economic modelling. This is discrepancy D-2 (§43).

A related asymmetry applies to Bribe reward tokens, whose decimals are unconstrained. A **low-decimal** reward token produces a proportionally larger whole-unit Fund liability at each carry-classification boundary, because one whole unit is economically larger. This cannot transfer pre-entry carry to a later signaler — the classification direction is always toward Fund — but it does mean low-decimal Bribe rewards leak more value to Fund at boundaries than high-decimal ones.

33.4 Overflow analysis

EXPRESSION	WIDEST INTERMEDIATE	SAFE BECAUSE
`mulDiv(E, 10^36, W)`	512-bit product	`Math.mulDiv` full-width intermediate
`mulDiv(w, Δ, 10^36)`	512-bit product	same
`mulDiv(bal, g, S)`	512-bit product	same
`(to − from) · rewardRate`	`uint256`	`rewardRate = ⌊S/604800⌋` ; `S` bounded by held balance
`emitted · 10^18` (Bribe)	`uint256`	guarded by `_requireScalableBalance`
`elapsed · slot.tps`	`uint256`	`tps ≤ 10^24` ; overflow needs ~10^52 s
`balanceOf(a) · (rpt − paid)`	`uint256`	bounded by conserved accounted balance

`Bribe._requireScalableBalance` rejects any `accountedRewardBalance` exceeding `type(uint256).max / 10^18` with `RewardScaleOverflow`. Since ADR 0035 this is defense in depth rather than the primary bound: the monotonic lifetime cap of §23.6 is checked first and is the guard with direct test coverage (`test_LifetimeRewardCapAcceptsTheExactLimitAndRejectsTheFirstExcessUnit`). No current test drives `RewardScaleOverflow` in isolation.

Solidity 0.8.26 provides checked arithmetic throughout; there are no `unchecked` blocks in the value-bearing paths.

34. MEV and timing analysis

34.1 Mining slot auctions

Slot replacement is a public descending-price opportunity with a deterministic, publicly computable price. It is inherently competitive and MEV-exposed:

- **Sniping.** Searchers will fill at the earliest profitable moment. This is the mechanism working as designed — competition compresses the clearing price toward fair value.
- **Front-running a replacement.** Mitigated by `expectedEpochId`: a competing fill increments `epochId` and reverts the victim's transaction rather than executing it at a worse price.
- **Zero-price sniping.** After one hour the price is zero, so an incumbent can be displaced for free. This is intrinsic to the design and is finding **M-02** (§39.7).

34.2 Strategy auctions

Identical structure. The buyer's protections are `expectedEpochId`, `deadline`, and `maximumPayment`. The auction has no reserve price, so a Strategy's inventory will clear at whatever the market bears, potentially at zero after full decay. Since the next epoch's starting price derives from the last clearing price, a coordinated series of cheap fills can depress subsequent starting prices; the `minimumPrice` floor bounds this, and recovery is geometric at `m_s`.

34.3 Stream restart timing

`notifyRevenue` requires `reward ≥ left`, so an actor wishing to force a restart must supply at least the remaining schedule. The influence that remains is genuine and accepted:

- An actor who *wants* faster emission can top up to restart at a higher rate.
- An actor who *wants* slower emission can top up with `reward ≈ left` late in a period, re-spreading the remainder over a fresh seven days.

Both cost real capital proportional to the remainder. There is no free griefing vector, but there is no manipulation-proof guarantee either.

34.4 Signal timing

P-1 (§19) eliminates same-transaction capture. It does **not** eliminate short-horizon strategic signaling: an actor observing an imminent large notification can allocate signal one block earlier and capture that interval's flow legitimately. Because there is no epoch, cooldown, or minimum duration, signal weight is fully fluid. This is a deliberate design decision, not an oversight.

34.5 Redemption timing

A redeemer's payout depends on `gbx.totalSupply()` at execution. Adversarial interleaving is bounded:

- Mining accrual is force-checkpointed (§25.3), so it cannot be timed against.
- Unburned Fund GBX inflates the denominator, so a redeemer benefits from burning it first — a permissionless action any redeemer can take in the same transaction bundle.
- Another redemption in the same block reduces both numerator and denominator consistently (I-9), so ordering does not create extractable advantage.

34.6 Checkpoint griefing

Because accrual is lazy and every mandatory loop is bounded, there is no unbounded-work griefing vector. The costs a griefer can impose are the fixed 8-token Bribe loop on a signaler entering or exiting. Fund redemption reads Mine's effective supply in constant time and performs no mining-slot loop.

35. Economic analysis

35.1 The mining market

A miner's expected return over a tenure of length `T` is:

```
E[return] = T · tps · price_GBX  +  Pr[replaced at price p > 0] · E[⌊0.8p⌋]  −  p_entry
```

Equilibrium properties:

- **`p_entry` is set by the descending auction**, so it converges toward the market's valuation of `T·tps·price_GBX` plus the option value of the handoff.
- **The handoff term is not guaranteed.** It is zero if no successor pays, and the price reaches zero after one hour.
- **Tenure length is endogenous**: a slot that is profitable to hold is also profitable to take, so `T` shortens as GBX price rises.

- **The multiplier `m` is a ratchet.** A hot slot escalates its own next starting price geometrically, damping the rate at which cheap tenures can be acquired in succession.

35.2 Issuance dynamics

Global issuance is `u_k` per second, halving through a schedule that terminates below cumulative `2H` (§12.5), after which it is permanently `u_∞`. Two consequences:

- **The tail is the long-run regime.** Almost all of the protocol's lifetime is spent at `u_∞`, not on the halving curve. Parameter selection should therefore weight the tail heavily.
- **Capacity expansion transiently over-issues** (M-01, §12.6). The excess is bounded by the incumbents' legacy rates and decays as tenures turn over.

35.3 Signal equilibrium

A signaler's return from allocating weight `w` to Strategy `s` over interval `[t₀, t₁]` is:

```
Bribe rewards to s over [t₀,t₁] · w / totalSupply(s)
```

Note what is **absent**: the signaler receives no share of the revenue they direct, and no share of the acquired asset. Revenue directed to `s` benefits *all GBX holders* via Fund backing, while Bribe rewards accrue only to signalers of `s`.

This creates the intended separation: **acquisition is a public good funded by protocol revenue; direction is a private good funded by whoever wants that direction.** The equilibrium is that signal flows toward Strategies whose Bribe yield is highest, and Bribe funders bid against each other for the protocol's acquisition capacity.

Failure mode. If no Strategy is bribed, rational signalers keep sGBX idle, `totalSignalWeight` falls toward zero, and stream emission during that interval becomes permanently unclaimable surplus (§17.5). The protocol continues to function but leaks revenue. Nothing in the design prevents this.

35.4 Redemption and backing

Backing per GBX is `balance_i / totalSupply` for each asset `i`. It rises when: assets are acquired, GBX is burned (Fund burns, LP-fee burns, redemptions with floored residue). It falls when: GBX is issued by mining. There is no onchain computation of aggregate backing, because that would require prices.

Redemption arbitrage. If the market price of GBX falls below the redeemable value of the basket, burning becomes profitable, which contracts supply and raises backing per remaining token. This is the design's only value-stabilization mechanism, and it is entirely emergent — no contract enforces or subsidizes it.

35.5 What the design does not attempt

The protocol maintains index **membership** — the set of Strategies registered in `Resonance` — but no index **methodology**. There are no target weights, no rebalancing, no drift correction, no diversification constraint, no risk budget, no performance fee, no management fee, no NAV, and no valuation. Fund's composition is a historical record of what signalers directed and buyers filled, plus whatever anyone donated; the proportions are an outcome, not a target.

36. Supported-token model

36.1 Support definition

A token is supported only if:

1. a requested transfer debits the sender by **exactly** the requested amount;
2. it credits the receiver by **exactly** the requested amount;
3. `balanceOf`, `approve`, `transfer`, `transferFrom` follow conventional semantics;
4. balances do not change asynchronously (no rebasing);
5. callbacks cannot bypass reentrancy guards or authorization.

36.2 Enforcement

Every value-moving path snapshots both balances and reverts on inexact movement:

CONTRACT	ERROR
`Mine`	`InexactTransfer`
`SignalGBX`	`InexactUnderlyingTransfer`
`Resonance`	`InexactRevenueTransfer`, `InexactRevenuePayout`
`Strategy`	`InexactPayment`, `InexactPayout`
`BribeRouter`	`InexactTransfer`
`Bribe`	`InexactRewardTransfer`, `InexactRewardPayout`
`Fund`	`InexactTransfer`
`LiquidityPosition`	`InexactTransfer`

This is fail-closed evidence, not support. It guarantees the protocol does not silently absorb loss from a fee-on-transfer or rebasing token; it does not make such a token usable, and it does not make an adversarial, upgradeable, pausable, or blocklisting token safe.

36.3 Explicit exclusions and accommodations

CATEGORY	HANDLING
Fee-on-transfer	Rejected by exact-delta checks in every path
Rebasing	Rejected in effect; asynchronous balance changes break the delta checks
ERC-777-style callbacks	Reentrancy-guarded; reentrancy regressions exist for hostile payment/reward tokens
Missing-return-value tokens	**Supported** via `SafeERC20`
Tokens reverting on zero approval	**Supported** — conditional zero-approval cleanup (finding E-04)
`SignalGBX` as payment or reward	**Forbidden** at registration (finding E-03)
Blocklisting tokens	Can block their own payout only; liability persists and is retryable

36.4 Failure isolation summary

FAILURE	BLAST RADIUS
Broken Strategy payment token	That Strategy's Fund liability; not signal exit, not other Strategies
Broken Bribe reward token	That token's claim only; other tokens claimable selectively
Broken Fund asset	Only redemptions that *select* it
Broken USDG	**Systemic** — all revenue, all auctions, all mining payments
Broken GBX	**Systemic** — impossible by construction; GBX is protocol-controlled

USDG is the single external token whose failure is systemic and unrecoverable.

37. External dependencies

DEPENDENCY	VERSION/SOURCE	TRUST REQUIRED	FAILURE IMPACT
OpenZeppelin ERC20/Permit/Votes	`@openzeppelin/contracts`	Library correctness	Systemic
OpenZeppelin `SafeERC20`, `Math`, `ReentrancyGuard`	same	Library correctness	Systemic
Uniswap v4 `IPositionManager`, `Actions`	`@uniswap/v4-periphery`	Correct fee accounting on zero-liquidity decrease	LP revenue only
Uniswap v4 core types	`@uniswap/v4-core`	Pool key semantics	LP revenue only
USDG	external, third-party	Solvency, no blocklist, no rebase, 6 decimals	**Systemic**
Strategy payment tokens	external, per Strategy	Standard ERC-20 behavior	Per-Strategy
Bribe reward tokens	external, per token	Standard ERC-20 behavior	Per-token
EIP-1153 transient storage	chain (Cancun)	`tstore` / `tload` availability	**Redemption fails**

Absent by design: no price oracle, no NAV computation, no entropy source, no keeper network, no cross-chain bridge, no off-chain signer, and no external upgrade authority.

Target chain. `README.md` names **Robinhood Chain** as the intended target. `packages/config/deployments` holds dated *candidate* files. No canonical USDG or Uniswap v4 address is resolved and no signed manifest clears them.

38. Threat model

38.1 Smart-contract bugs

Exposure. The full protocol; nothing is upgradeable. **Mitigation.** Small surface (3,532 lines of Solidity across 19 source files), immutability, exact-delta checks, reentrancy guards, extensive fuzz and stateful-invariant coverage (§40). **Residual. No independent audit has been performed.** A discovered bug cannot be patched, paused, or worked around. This is the single largest unmitigated risk in the system.

38.2 The `Resonance` owner (findings G-01, G-03, open)

The core's threat model for administration is exactly one address. Whoever holds `Resonance.owner()` can add Strategies, kill any Strategy except the final live one, register Bribe reward tokens up to the eight-token cap, transfer ownership onward, or renounce it. They **cannot** drain Fund, mint GBX, alter mining economics, reprice incumbents, or move the liquidity position — those surfaces are ownerless or immutable (§9.3). They can move the signaler share within `[0, MAX_BRIBE_BPS]`, which is an economic lever rather than a custody one: it never reaches an already-classified liability and never lowers Fund's cumulative share below 80%. Killing a Strategy is irreversible, making it the highest-impact capture target; renouncing ownership is equally irreversible and permanently freezes Strategy membership.

Because ADR 0034 removed the in-repository Governor and Timelock, the core supplies **no** mitigation of its own: no proposal filter, no quorum, no voting period, no execution delay, no cancellation, and no observable pending state. An owner call takes effect in the transaction that makes it. Every capture, collusion, and liveness question therefore transfers wholesale to the external governance system that has not been selected, and must be re-analyzed against that system's exact release rather than against this repository.

Two properties of sGBX will shape that analysis. First, because no sGBX can be idle (§13.4), `getPastTotalSupply` measures economically active weight only — an external system using it as a quorum denominator is not diluted by parked receipts, and the former undelegated-supply deadlock concern does not arise from the token. Second, and less comfortably, checkpoints survive withdrawal.

38.3 Short-duration voting weight (finding G-01)

sGBX has no staking or withdrawal lock, and its checkpoints record historical block balances permanently. An account may acquire or **borrow** GBX, signal it, allow the snapshot block to pass, and withdraw immediately afterwards, bearing only borrowing cost and price risk for a few blocks while retaining the recorded weight.

Nothing in the core acts on that weight today, so this is not presently exploitable. It becomes exploitable exactly when an external system is attached that reads historical balances, and its severity then depends on that system's snapshot-to-vote spacing and its proposal threshold. Combined with the irreversibility of `killStrategy`, an inadequately spaced integration would permit a low-cost, permanent hostile action. This is why snapshot compatibility is an explicit item in the ADR 0034 selection requirements (§15.4).

38.4 Oracle and external-price assumptions

None exist. The protocol never reads a price. This eliminates oracle manipulation as an attack class entirely. The substituted risk is that auction clearing prices may be poor if the auction is thin, uncompetitive, or filled at full decay.

38.5 Auction manipulation

- **Depressing Strategy prices.** A coordinated actor filling late repeatedly depresses subsequent starting prices, bounded below by `minimumPrice` with geometric recovery.
- **Inflating Mine prices.** Filling early repeatedly escalates starting prices geometrically at `m`, potentially pricing out honest miners — but each escalation costs the manipulator the full payment, 80% of which goes to the displaced party.
- **Inventory timing.** `Strategy.buy` distributes before reading inventory (§19.1), so a buyer always receives everything released through the execution timestamp. They cannot be shortchanged by a stale balance, nor can they acquire inventory that postdates their transaction.

38.6 MEV and transaction ordering

See §34. Mining and Strategy auctions are inherently MEV-exposed; `expectedEpochId`, `deadline`, and price caps convert ordering risk into transaction failure rather than economic loss.

38.7 Reward timing

Stream restarts are influenceable at real capital cost (§34.3). Signal is fully fluid across blocks (§34.4). Neither is manipulation-proof; both are bounded by requiring the manipulator to commit capital proportional to the effect.

38.8 Rounding extraction

Analysis. Rows 7 and 8 of §33.1 discard value rather than assigning it, so there is no party who *receives* the rounding residue and therefore no direct extraction target. An attacker could in principle maximize *waste* by forcing maximally-frequent checkpoints at maximally-adverse weights, but this destroys value rather than capturing it and costs the attacker gas. Bribe residues are carried exactly and classified to Fund, so they cannot be extracted by a later entrant (S-11, S-12).

Verified negative results: `test_NewSignalerCannotReceivePreEntryRewardCarry`, `test_NewStrategySignalCannotReceivePreEntryRoundedSurplus`, `test_ZeroSignalElapsedRevenueBecomesSurplusAndCannotBeCapturedLater`.

38.9 Malicious and nonstandard tokens

Covered in §36. Blast radius is per-token except for USDG.

38.10 Fee-on-transfer and rebasing assets

Rejected by exact-delta checks in every ingress and egress path. A token that *becomes* fee-on-transfer after registration (an upgradeable token enabling a fee) causes its own payouts to revert; the liability persists and becomes claimable again if the fee is disabled (`test_FeeEnabledAfterNotificationRollsBackTheClaimAndCanRetry`).

38.11 Proxy upgrades in external tokens

Unmitigated and unmitigable. Any Strategy payment token, Bribe reward token, Fund asset, or USDG may be a proxy whose implementation changes to add a blocklist, a fee, a pause, or an arbitrary drain. The protocol has no allow-list to remove them from and no rescue path. A redeemer's only defense is omitting the asset.

38.12 Chain censorship and reorganization

Auction fills, mining replacements, and redemptions are ordinary transactions subject to censorship and reorg. A reorg can undo a fill, changing who holds a slot or who received an auction's inventory. Nothing in the protocol assumes finality beyond ordinary chain guarantees. Prolonged censorship of `route` or a slot handoff delays settlement but does not destroy entitlements, since all accrual is time-based and lazily settled.

38.13 Immutable-deployment mistakes

See §28.4. Eight categories of permanent, unrepairable error. This is finding **M-03**, an open High release gate.

38.14 Compromised frontend or indexer

The contracts are permissionless and directly callable, so a compromised interface cannot steal funds directly. It can, however:

- mislead a redeemer into omitting valuable assets (permanently forfeited);

- mislead the final signaler of a dead Strategy into exiting and abandoning rewards;
- present raw balances as inventory, misleading auction participants;
- present the 80% mining handoff as guaranteed;
- present unsolicited Fund tokens as protocol-endorsed holdings.

Every one of these is a **permanent** loss to the misled user. Interface correctness is therefore a genuine security boundary, not merely a usability concern.

38.15 Incorrect deployment parameters

Finding **M-04**, open High release gate. The Mine's initial rate, halving amount, tail rate, price multiplier, and minimum price are unselected. Wrong values produce unsafe or unusable economics even with perfectly correct Solidity. The external governance system's own parameters are equally unselected, but they are outside this repository (§15.4).

38.16 Loss of keys

The deployment coordinator's key is critical **only during setup**, and only until ownership is handed to the external executor (§27.3). Mine, Fund, and LiquidityPosition are ownerless, so no key loss affects them. Resonance's owner key is exactly as critical as the external system that holds it, which is unselected; a lost owner key permanently freezes Strategy membership in the same way `renounceOwnership` would. Loss of a *user's* key loses that user's GBX and any allocated stake permanently, with no recovery mechanism.

38.17 Legal and regulatory risk

Unresolved in every respect. Additionally, upstream code provenance and license reconciliation are explicit release blockers (§41.5): the protocol adapts pinned give.fun, Liquid Signal Governance, and Farplace MineRig code, with a transitive **GPL-2.0-or-later** ancestor in the auction lineage and unidentified Synthetix and Solidly ancestors, while the repository declares BUSL-1.1 at the root and MIT per file.

39. Residual risks

Risks that remain after all implemented mitigations, in descending severity.

39.1 No independent audit

No third party has reviewed this code at any version. Internal testing (§40) is engineering evidence and is not a substitute. **Open.**

39.2 Unrepairable defects

Immutability means any defect — in code, parameters, or deployment — is permanent. **Accepted by ADR 0016/0017.**

39.3 Unselected external governance

Finding **G-03**. The core has no governance of its own, and the system that will own `Resonance` has not been chosen. Until it is, the protocol's capture resistance, liveness, delay, and accountability properties are undefined rather than weak. A deployment that skipped the handoff would ship an ordinary admin key. **Open release gate.**

39.4 Historical voting weight outlives the position

Finding **G-01**. sGBX checkpoints survive withdrawal, so any external system reading historical balances must space its snapshot and voting window deliberately. Not exploitable in the core, which reads no checkpoints. **Open integration gate.**

39.5 Unbounded Resonance surplus

Findings **A-02/A-09**. Rounding floors, zero-weight intervals, and donations accumulate permanently with no recovery path and no lifetime bound. **Accepted by ADR 0029.**

39.6 Unbounded reward abandonment in dead Strategies

Finding **BR-1**. A final signaler's exit can strand a complete unvested stream. **Accepted by ADR 0028.**

39.7 Miner rollover and zero-price replacement

Finding **M-02**. The 80% handoff is contingent, not guaranteed, and after one hour a successor pays nothing. **Accepted by ADR 0024.**

39.8 Transitional over-issuance after a halving

Finding **M-01**. Aggregate issuance can exceed the undivided global rate until legacy tenures turn over. **Accepted by ADR 0033.**

39.9 Unselected economic parameters

Finding **M-04**. **Open release gate.**

39.10 Lookalike dependencies at deployment

Findings **M-03/E-02**. Reciprocal checks prove consistency, not honesty. **Open release gate.**

39.11 Unharvested liquidity fees

No caller bounty exists, so fees may accrue indefinitely unharvested. Value is not lost — it remains claimable by any future harvester — but revenue timing is unpredictable.

39.12 Forfeited redemption assets

A redeemer who omits an asset permanently forfeits their claim to it. Interface error here is unrecoverable.

39.13 Unsolicited Fund assets

Any ERC-20 can become Fund backing without review. Malicious tokens in the Fund cannot harm redeemers who omit them, but can mislead observers about the treasury's composition and quality.

39.14 USDG issuer risk

A single external stablecoin is the substrate for all revenue, all auctions, and all mining payments. Its failure is systemic and unrecoverable.

39.15 Legal and provenance

§38.17. **Open release blocker.**

40. Testing and verification evidence

> **Every figure in §40.1 was produced by executing the suites against the tree at commit `dc67d7c4d634097fa6e285-fa33ce964d591d2bd2` on 20 August 2026.** Figures from earlier commits are segregated into §40.4 and labelled with their own commit and date. Static analysis, mutation testing, and external fuzzing were **not** re-run at this commit; §40.4 records their pinned results and their staleness explicitly. **None of this constitutes formal proof or an independent audit.**

40.1 Verified at the reviewed commit

Command: `forge test --summary`

RESULT	VALUE
Suites	21
Passed	**329**
Failed	0
Skipped	0

SUITE	PASSED	SUITE	PASSED
`AdversarialTest`	18	`LiquidityPositionDeepTest`	18
`ArchitectureReconciliationRegressionTest`	4	`MineTest`	21
`BribeTest`	32	`ResonanceTest`	35
`BribeRetirementRiskTest`	1	`BribeRouterTest`	15
`BribeRewardFlowTest`	10	`ResonanceRouterTest`	8
`CarryReallocationTest`	4	`SignalGBXTest`	23
`FactoriesTest`	8	`SignalGasTest`	4
`FundTest`	26	`StartingPointTest`	15
`GBXTest`	10	`StrategyTest`	40
`HistoricalBribeDifferentialTest`	3	`USDGFlowTest`	5
`ProtocolInvariantsTest`	29		

This figure **matches `FINDINGS.md` at this commit exactly.** `ProtocolGovernorTest` and its 11 tests were removed with the Governor itself (ADR 0034). The ADR 0035 lifetime reward cap added coverage to `BribeTest`, `BribeRewardFlowTest`, and `BribeRouterTest`; ADR 0034's checkpoint-retention behavior added coverage to `SignalGBXTest`.

Command: `FOUNDRY_PROFILE=integration forge test --summary`

RESULT	VALUE
Suites	2
Passed	**18**
Failed	0
Skipped	0

`CampaignHarnessTest` 7, `LiquidityFeeHarvestTest` 11.

40.2 Campaign configuration

From `packages/contracts/foundry.toml` at this commit:

PROFILE	FUZZ RUNS	INVARIANT RUNS	INVARIANT DEPTH	`FAIL_ON_REVERT`
`default`	10,000	1,000	500	true
`ci`	10,000	1,000	500	true
`nightly`	100,000	10,000	1,000	true
`integration`	256	—	—	—

Compiler: Solidity 0.8.26, Cancun, optimizer enabled at 10,000 runs, no metadata hash. Foundry 1.7.1 (`4072e48705af9d93e3c0f6e29e93b5e9a40caed8`).

> The `nightly` profile was **not** executed for this document and is not reported as passing.

40.3 Evidence by method

Unit and negative testing. 329 default-profile tests spanning constructor validation, authorization, degenerate arguments, revert paths, and behavioral regressions for every accepted finding.

Property-based fuzzing. 22 `testFuzz_` properties in the default profile at 10,000 runs each — **220,000 configured fuzz cases**. Properties cover supply reconciliation (I-1), receipt collateralization (I-3), signal-backing (I-4), 90/10 frequency-independence (F-21), routing conservation (I-11), redemption pro-rata exactness and monotone backing (I-9), price-curve exactness and monotonicity, next-price bounds, Bribe solvency (I-8), and Resonance non-overpayment (I-7).

Stateful invariant testing. `ProtocolInvariantsTest` contains **27** `invariant_` entries — 26 asserting properties plus `invariant_CallSummary`, which reports selector coverage — and two deterministic regressions (`test_EveryHandlerActionIsReachable` , `test_DynamicallyAddedStrategyEntersEveryHarnessPath`), for the 29 tests reported above. Each invariant entry runs at 1,000 runs × depth 500 with `fail_on_revert = true` — **500,000 calls per entry**, **13,500,000 aggregate state-machine transitions**. This run reached all **29 handler selectors** between 16,989 and 17,470 times each, with **zero handler reverts and zero discards**. The reachability regression exists specifically to prevent a permanently short-circuited handler action from producing false confidence.

Integration testing against real Uniswap v4. `LiquidityFeeHarvestTest` (11 tests) deploys real Uniswap v4 and exercises the zero-liquidity `DECREASE_LIQUIDITY` fee-collection path, principal preservation across repeated harvests, harvesting after price leaves the range, and atomic rollback of routing failure. This suite lives in a separate profile because Uniswap's `PositionManager` exceeds EIP-170 when built at this project's optimizer settings.

Randomized action-sequence campaign. `CampaignHarnessTest` (7 tests) wires the complete protocol graph from a single constructor and runs `testFuzz_RandomActionSequencesPreserveEveryProperty` over 256 randomized 12-action sequences, asserting every property after each action.

Architecture-reconciliation regressions. `ArchitectureReconciliationRegressionTest` (4 tests) is the direct evidence for the ADR 0031/0032 changes: it asserts the removed idle-receipt selectors are absent from **deployed runtime**, that `signal` atomically custodies, mints, delegates, and mirrors the paired Bribe, and that ten one-raw-unit payments classify to exactly Fund 9 / Bribe 1.

Differential-model evidence. `packages/simulations` contains independent TypeScript and Python economic models whose committed JSON fixtures and SVG charts are reproducibility evidence, checked by `pnpm simulations:fixtures:check`. The fixtures independently reproduce the mining price curve, the 80/20 mining split, staggered fixed-slot pre/post-halving tenures, the halving schedule, the tail, the Strategy auction curve, the supply identity, and effective-supply redemption dilution.

Gas bounding. `test_MaximumRewardTokenGasStaysFarBelowABlock`, `test_RewardTokenGasSlopeIsRecordedAndBounded`, `test_ScalarSignalEntryAndExitRemainCheapInTheShippedConfiguration`, and `test_FixedLiabilityAndGovernanceGasIsRecorded` bound the mandatory loops of §32 (L-10).

Reward-cap regressions (ADR 0035). `test_LifetimeRewardCapAcceptsTheExactLimitAndRejectsTheFirstExcessUnit`, `test_LifetimeRewardCapStillBlocksAfterTheMaximumWasClaimed`, `test_LifetimeRewardCapFailurePreservesRouterStateAndFundSettlement`, and `test_KilledStrategyExitRemainsLiveAfterRewardLifetimeCapIsConsumed` establish that the monotonic per-token counter is not reset by claims, Fund payment, stream completion, or Strategy death, that a rejected notification leaves the caller's tokens and every liability untouched, and that signal exit stays available after the cap is consumed (§23, L-9).

Checkpoint-retention regressions (ADR 0034). `test_HistoricalVotingCheckpointsSurviveImmediateSignalWithdrawal` and `test_LaterSignalPreservesExplicitDelegateAndSelfDelegatesAgainAfterZeroDelegation` establish the two checkpoint properties §15.3 relies on, and `test_DirectDonationIsSurplusAndCreatesNoSignalVotesOrWithdrawalEntitlement` establishes that a direct GBX donation creates no voting weight.

Static analysis, external fuzzing, and mutation testing — pinned to an earlier tree. These three campaigns were last executed before ADR 0034 removed the Governor and ADR 0035 added the Bribe lifetime cap. `FINDINGS.md` states explicitly that their results "predate ADRs 0034 and 0035 and remain historical engineering evidence." They are reported in §40.4 rather than here, and **must not be cited as current-tree evidence**. Re-running them at this commit is an open task.

Adjacent gates that do pass at this commit. Hardhat bytecode parity, the SDK test suite and ABI check, the subgraph build and tests, the independent TypeScript and Python simulation fixtures, contract and SDK documentation generation, formatting, linting, type checking, and the workspace build.

40.4 Historical evidence — explicitly not current

`packages/contracts/audit/AUDIT-BASELINE.md` and `packages/contracts/audit/TEST-CAMPAIGN.md` both carry "Historical evidence only" banners. They review commit **`54e3f2c3ce1de25aea4da2f21fab27804a3bfa84`** (2026-08-09), which **predates** the ADR 0024 Mine redesign and the ADR 0029/0030/0031/0032 changes. Their reported figures — including **340 default Foundry tests** — describe a superseded contract graph and must not be read as current.

At this commit the register and the executed suites agree exactly: `FINDINGS.md` reports **329** default and **18** integration tests, and both were reproduced by running the suites (§40.1). See discrepancy D-4 for the history of this figure.

Static analysis, external fuzzing, and mutation results are also historical. Their last recorded pass covered a tree predating ADR 0034 and ADR 0035: pinned **Slither 0.11.5, Aderyn 0.6.8, Semgrep 1.162.0, Gitleaks 8.30.1** with a register of 177 accepted source findings across 28 detector classes and zero raw Semgrep/Gitleaks findings; native **Medusa 1.5.1** at **101,602 calls** with zero failures across 65 surfaces; **Echidna 2.3.2** at **100,213 calls** with all **25 properties** passing; and a focused **43-mutant** campaign that killed every mutant. Mythril 0.24.8 was incompatible with constructor-resolved immutable/Cancun runtimes and was never a proof. Because the Governor removal deleted a contract and the reward cap added a new state variable and revert path, these figures describe a different contract graph than the one this document specifies.

40.5 Verification methods absent

METHOD	STATUS AT THIS COMMIT
Independent external audit	**Not performed**
Static analysis at this commit	**Not re-run** since ADR 0034 and ADR 0035 (§40.4)
External fuzzing at this commit	**Not re-run** since ADR 0034 and ADR 0035 (§40.4)
Mutation testing at this commit	**Not re-run** since ADR 0034 and ADR 0035 (§40.4)
Symbolic execution	Not performed (Mythril incompatible with this runtime)
Formal verification	Not performed
Second external-fuzzer seed	Not performed
Fork validation	Not completed; no RPC capability and block pin recorded
Monitored testnet rehearsal	Not performed
Reviewed production parameters	Not selected
Release review	Not performed
Signed deployment manifest	Does not exist

> A skipped fork run is not a pass. Fork results count only when the exact RPC capability and block pin are recorded.

41. Deployment status

41.1 Deployment

The protocol is not deployed on any network. No signed deployment manifest exists for this repository state. `packages/config/deployments` contains dated *candidates* and *policy* files (for example `robinhood-mainnet-wrapped-btc.2026-08-02.candidate.json`, `provisional-mainnet-bytecode-hashes.2026-08-01.json`), none of which is a cleared canonical address.

`docs/DEPLOYMENT.md` is explicitly “an unexecuted development outline, not a deployment manifest or release authorization,” and no script in the repository is authorized to broadcast it.

41.2 Review

Internal engineering review only. Dispositions are recorded in `packages/contracts/audit/FINDINGS.md` (2026-08-16, with governance and Bribe-cap dispositions reconciled 2026-08-19 for ADRs 0034 and 0035), with campaign-specific findings in `packages/contracts/audit/SIGNAL-RESONANCE-FINDINGS.md`.

41.3 Open release gates

FINDING	SEVERITY	GATE
M-03	High	Immutable bindings cannot detect a malicious lookalike. Requires signed manifest, exact runtime code hashes, constructor arguments, and receipts.
M-04	High	Mine's initial rate, halving amount, tail rate, price multiplier, and minimum price are unselected and unmodelled.
G-03	High	The external governance system that will own `Resonance` is unselected; its voting, delegation, permission, and delay semantics are unreviewed.
G-01	High	sGBX checkpoints survive withdrawal; the selected external system's snapshot-to-vote spacing requires independent review of the capture model.
E-02	High	Materially reduced by reciprocal identity checks, but codehash, parameter, and manifest review remains external.

41.4 Accepted findings (not gates)

A-02, **A-09** (Resonance half) — accepted by ADR 0029. **BR-1** — accepted by ADR 0028. **M-01**, **M-02** — accepted by ADR 0024. **G-02** — accepted by ADR 0030. **A-08** — liveness resolved; bounded cost retained. **SR-001** (idle sGBX and duplicate signal ledgers) and **SR-002** (superseded 100%-Fund classification) — both High, both fixed locally by ADR 0031 and ADR 0032 respectively and covered by named regressions.

41.5 Legal and provenance

An **unresolved release blocker**. `docs/LEGAL-PROVENANCE-BLOCKER.md` records that the active contracts adapt pinned give.fun `ef6ee14a…`, Liquid Signal Governance `14b5fbbb…`, and Farplace MineRig `8cf74230…`; that `Strategy`'s descending-price shape has a transitive Euler Fee Flow ancestor at `3bee858a…` whose reviewed file is **GPL-2.0-or-later**; that stated Synthetix and Solidly ancestors lack exact repository, commit, and path; that `LiquidityPosition` cites a TokenJar concept with no recorded provenance; and that the repository declares BUSL-1.1 at the root while every active Solidity file declares MIT. No clean-room, compatibility, relicensing, or separate-permission claim is made.

41.6 Status language

The following terms are **not** applicable to this protocol at this commit and are not used in this document except to deny them: *audited*, *safe*, *verified*, *launched*, *live*, *production-ready*, *trustless*, *risk-free*, *fully decentralized*, *community-owned*, *guaranteed yield*.

Terms that **are** supported by evidence: *immutable* (no upgrade path exists in `packages/contracts/src`), *governance-minimized* (three selector-bounded actions), *ownerless* (of `Mine`, `Fund`, and `LiquidityPosition`), *permissionless* (of the named user-facing operations), and *non-transferable* (of sGBX).

42. Contract reference

CONTRACT	PATH (UNDER `PACKAGES/CONTRACTS/SRC`)	LINES	INHERITS	KEY CONSTANTS
`GBX`	`core/GBX.sol`	95	`ERC20`, `ERC20Permit`	`GENESIS_LIQUIDITY_ALLOCATION = 20_000_000 ether`
`Mine`	`core/Mine.sol`	375	`ReentrancyGuard`	`BPS 10_000`, `PREVIOUS_MINER_BPS 8_000`, `PRICE_DECAY_PERIOD 1 hours`, `SLOT_COUNT 16`, `MIN_TAIL_TPS 16`, `MAX_INITIAL_TPS 1e24`
`SignalGBX`	`core/SignalGBX.sol`	195	`ERC20`, `ERC20Votes`, `ReentrancyGuard`, `Ownable`	—
`Resonance`	`core/Resonance.sol`	459	`ReentrancyGuard`, `Ownable`	`DURATION 7 days`, `REWARD_PRECISION 1e36`
`ResonanceRouter`	`core/ResonanceRouter.sol`	83	`IResonanceRouter`, `ReentrancyGuard`	—
`Strategy`	`core/Strategy.sol`	238	`ReentrancyGuard`	`MIN_EPOCH_DURATION 1 hours`, `MAX_EPOCH_DURATION 365 days`, `PRICE_SCALE 1e18`, `ABSOLUTE_MINIMUM_PRICE 1e6`
`StrategyFactory`	`core/StrategyFactory.sol`	82	`Ownable`	—
`Bribe`	`core/Bribe.sol`	747	`ReentrancyGuard`	`REWARD_DURATION 7 days`, `REWARD_PRECISION 1e18`, `MAX_REWARD_TOKENS 8`, `MAX_LIFETIME_REWARD_AMOUNT` $\lfloor(2^{256}-1)/1e18\rfloor$
`BribeFactory`	`core/BribeFactory.sol`	65	`Ownable`	—
`BribeRouter`	`core/BribeRouter.sol`	203	`ReentrancyGuard`	`BPS 10_000`, `FUND_BPS 9_000`, `BRIBE_BPS 1_000`
`Fund`	`core/Fund.sol`	193	`ReentrancyGuard`	`REDEMPTION_NAMESPACE`
`LiquidityPosition`	`core/LiquidityPosition.sol`	342	`IERC721Receiver`, `ReentrancyGuard`	—

Interfaces: `ICoreResonance` (54), `IResonanceIdentity` (25), `IBribe` (21), `IFund` (16), `IMine` (15), `IResonanceRouter` (12). `ISignalGBXAllocation` was **deleted** by ADR 0031.

Total protocol Solidity: 3,220 lines across 18 files — 12 core contracts and 6 interfaces. There is no `governance/` source tree: ADR 0034 deleted `ProtocolGovernor.sol` (246 lines) along with its tests, ABIs, SDK lifecycle helpers, and subgraph data sources.

42.1 Permissionless entry points

`Mine.mine`, `Mine.claim`; `ResonanceRouter.route`; `Resonance.distribute`; `SignalGBX.signal` / `signalWithPermit` / `moveSignal` / `withdrawSignal`; `Strategy.buy`; `Bribe.notifyRewardAmount` / `claimReward` / `claimRewards` / `payFundReward`; `BribeRouter.payFundPayment` / `notifyBribeReward`;

`Fund.burnGBX` / `redeem` ; `LiquidityPosition.harvestFees` ; `GBX.burn` . `SignalGBX.delegate` / `delegateBySig` are permissionless but read by nothing in the core (§15.3).

42.2 Restricted entry points

FUNCTION	CALLER
`GBX.mint`	`Mine` (locked)
`GBX.setMinter`	current `minter` , once
`Resonance.addSignalFor` / `removeSignalFor` / `moveSignalFor`	`SignalGBX`
`Resonance.notifyRevenue`	`ResonanceRouter`
`Resonance.addStrategy` / `killStrategy` / `addBribeReward`	owner (**unselected**)
`Resonance.setResonanceRouter`	owner, once
`Bribe.deposit` / `withdraw` / `addRewardToken`	`Resonance`
`BribeRouter.routePayment`	its immutable `Strategy`
`StrategyFactory.createStrategy` , `BribeFactory.createBribe`	bound `Resonance`
`SignalGBX.setResonance` , factories' `setResonance`	owner, once

43. References and provenance

43.1 Source of truth

ARTIFACT	PATH
Solidity implementation	`packages/contracts/src`
Executable specification suite	`packages/contracts/test/minimal/StartingPoint.t.sol`
Internal fact registry	`docs/facts/gumball-6900-facts.md`
Finding register	`packages/contracts/audit/FINDINGS.md`
Economic fixtures	`packages/simulations/fixtures/economic-scenarios.json`

43.2 Authoritative ADRs

ADR 0017 (ownerless Fund and LiquidityPosition, no successor) · ADR 0022 (fixed-principal LP fee routing) · ADR 0024 (immutable multislot Mine; its GBX-ERC20Votes statement superseded by ADR 0030) · ADR 0027 (Bribe carry boundaries) · ADR 0028 (closed Bribe pools after Strategy death) · ADR 0029 (Bribe-based Resonance; signal entrypoints superseded by 0030 then 0031, kill-final-Strategy by 0031, 100%-Fund by 0032, intended Timelock owner by 0034) · ADR 0030 (non-transferable ERC20Votes sGBX; its `ProtocolGovernor` , Timelock, selector-filter, and cancellation decisions superseded by ADR 0034, and its idle-sGBX and `allocatedBalance` decisions by ADR 0031) · **ADR 0031 (mandatory signal-backed SignalGBX) · ADR 0032 (fixed 90/10 acquired-asset settlement) · ADR 0033 (fixed sixteen Mine slots and constant-time pending emission) · ADR 0034 (external governance ownership) · ADR 0035 (Bribe lifetime reward cap)**.

43.3 Superseded ADRs excluded from this document

Fully superseded: ADR 0018 (auto-compounding LP → ADR 0022) · **ADR 0021 (uniform 100%-Fund Strategy settlement → ADR 0032)** · ADR 0023 (fixed supply and Fundraiser reserve → ADR 0024) · ADR 0025 (global revenue stream → ADR 0026) · ADR 0026 (exact successor stream → ADR 0029).

Partially superseded, historical context only: ADR 0013 (acquisition splits, buyback, proposer model) · ADR 0014 (mint authority, distribution, fee routing) · ADR 0015 (whole-account actions, public coordination surface) · ADR 0016 (terminology; "management fee" means the bounded acquisition auction) · ADR 0019 (Resonance batch APIs, direct signal entrypoints, idle allocation and standalone exit) · ADR 0020 (Resonance carry, donation synchronization, Strategy routing — the last superseded by ADR 0021 and then ADR 0032).

43.4 Upstream lineage

Pinned give.fun `ef6ee14a454432210d13e312d0ef825f670bd79d`; Liquid Signal Governance `14b5f-bbbe1945f2e6501f84976e5f12b39fb227a`; Farplace MineRig `8cf7423016b108e7bd8d7854c14e0ac6585bb935`; transitive Euler Fee Flow `3bee858a1568d1313f37d615953f83391a897866` (**GPL-2.0-or-later**); stated Synthetix `StakingRewards` and Solidly ancestors, **exact sources unresolved**. Full per-file mapping and SHA-256 evidence: `NOTICE` and `docs/LEGAL-PROVENANCE-BLOCKER.md`. **See §41.5 — this is an open release blocker.**

43.5 Recorded discrepancies

D-1 — "Reverse Dutch auction" naming. `AGENTS.md`, `docs/EMISSIONS.md`, and the contracts' own NatSpec use "reverse Dutch." Mechanically both `Mine` and `Strategy` implement a **descending-price** auction, conventionally a plain Dutch auction; "reverse Dutch" follows the Euler Fee Flow lineage. *Resolution:* the repository's term is retained and the mechanics are always stated explicitly.

D-2 — Six-decimal USDG is an assumption, not a constant. No contract reads or validates `usdg.decimals()`. The only in-repository evidence for 6 is the test fixture (`ProtocolFixture.sol:65`, `MockERC20("Global Dollar", "USDG", 6)`) and the simulation fixtures. *Resolution:* stated throughout as a deployment property of the intended USDG, never as a contract guarantee. See §33.3.

D-3 — Stale `Fundraiser` artifact. `packages/contracts/artifacts/hardhat/src/core/Fundraiser.sol` exists as compiler output with no corresponding source; the design was superseded by ADR 0024. *Resolution:* excluded entirely. Regenerating Hardhat artifacts would clear it; generated artifacts were out of scope for this documentation work.

D-4 — Test-count drift (now resolved). `TEST-CAMPAIGN.md` reports 340 at commit `54e3f2c3` (2026-08-09); `FINDINGS.md` reported 322 at `281e601` while the actual figure there was 339; the pre-ADR-0034 campaign recorded 335 default and 17 integration. **At this commit the register and the executed suites agree exactly at 329 default and 18 integration.** *Resolution:* only figures verified by running the suites are reported as current; historical figures carry their own commit and date. See §40.1 and §40.4.

D-5 — Ownership closure is procedural. `docs/ACCESS_CONTROL.md`, `docs/INVARIANTS.md`, and `docs/TRUST_ASSUMPTIONS.md` state ownership and administrator conditions as invariants. **No Solidity enforces them**; they are deployment steps 9–10. Under ADR 0034 the specific obligation changed — from Timelock role closure to transferring `Resonance` to a reviewed external executor and proving the setup owner retains nothing — but its procedural character did not. *Resolution:* described throughout as deployment obligations requiring signed evidence. See §27.2 and §27.3.

D-7 — Documents were re-derived after a mid-work protocol change. An earlier version of this whitepaper, the one-pager, the article, and the fact registry described commit `281e601`. That commit was superseded by ADR 0031 (mandatory signal-backed SignalGBX) and ADR 0032 (fixed 90/10 acquired-asset settlement) before publication. The superseded drafts asserted that 100% of every auction payment became a Fund liability, that no auction proceeds fund Bribes, that idle sGBX could vote without earning, and that a standalone staking surface existed — **all four are now false**. *Resolution:* every affected claim was re-derived against `95ed60e` and the suites re-run. Any copy of these documents citing `281e601` should be discarded.

D-8 — Governance was removed after the previous edition was written. Edition 1.x of this whitepaper, the one-pager, the article, and the fact registry documented an in-repository `ProtocolGovernor` and `TimelockController`, including quorum formula F-8, the selector-bounded proposal filter, disabled-surface behavior, and the uncancellable queued-operation finding G-02. ADR 0034 deleted both contracts. Those sections previously carried "superseded" banners while retaining present-tense claims; in this edition they are **replaced** by §15 and §27, which specify the owner-gated surface that actually exists and state explicitly which guarantees no longer have any source of enforcement. Findings **G-02** and the local-parameter form of **G-03** are superseded by removal, not proven safe. Any copy of these documents describing a Governor or Timelock as current is stale.

D-6 — "Index protocol" framing. `README.md` calls the protocol an "onchain index protocol," while `AGENTS.md:72` uses the narrower and more precise formulation: "Official protocol/index membership is represented by Strategies registered in Resonance, not by a Fund asset list." *Resolution:* this document adopts the `AGENTS.md` sense throughout. Registered Strategies **are** index membership — the curated list of target assets — but the protocol supplies no index methodology: no weights, rebalancing, drift correction, reconstitution, or NAV (§4 N1, §35.5). Membership is never inferred from a Fund balance, since Fund accepts unsolicited transfers without review (§24.2).

43.6 Companion documents

- GUM BALL 6900 at a Glance — one-page summary
- How GUM BALL 6900 Turns Community Conviction Into an Onchain Portfolio — plain-English explanation
- Internal fact registry — per-claim evidence with source, tests, and caveats

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Reviewed commit: `95ed60efe333d875f7a66da7853eebdf5384e956`*. Not deployed. Not independently audited. Not approved for user funds.*